



HYPOTHESIS TESTING

8.1 INTRODUCTION

As in the previous chapter, let us suppose that a random sample from a population distribution, specified except for a vector of unknown parameters, is to be observed. However, rather than wishing to explicitly estimate the unknown parameters, let us now suppose that we are primarily concerned with using the resulting sample to test some particular hypothesis concerning them. As an illustration, suppose that a construction firm has just purchased a large supply of cables that have been guaranteed to have an average breaking strength of at least 7,000 psi. To verify this claim, the firm has decided to take a random sample of 10 of these cables to determine their breaking strengths. They will then use the result of this experiment to ascertain whether or not they accept the cable manufacturer's hypothesis that the population mean is at least 7,000 pounds per square inch.

A statistical hypothesis is usually a statement about a set of parameters of a population distribution. It is called a hypothesis because it is not known whether or not it is true. A primary problem is to develop a procedure for determining whether or not the values of a random sample from this population are consistent with the hypothesis. For instance, consider a particular normally distributed population having an unknown mean value θ and known variance 1. The statement " θ is less than 1" is a statistical hypothesis that we could try to test by observing a random sample from this population. If the random sample is deemed to be consistent with the hypothesis under consideration, we say that the hypothesis has been "accepted"; otherwise we say that it has been "rejected."

Note that in accepting a given hypothesis we are not actually claiming that it is true but rather we are saying that the resulting data appear to be consistent with it. For instance, in the case of a normal $(\theta, 1)$ population, if a resulting sample of size 10 has an average value of 1.25, then although such a result cannot be regarded as being evidence in favor of the hypothesis " $\theta < 1$," it is not inconsistent with this hypothesis, which would thus be accepted. On the other hand, if the sample of size 10 has an average value of 3, then even though a sample value that large is possible when $\theta < 1$, it is so unlikely that it seems inconsistent with this hypothesis, which would thus be rejected.

8.2 SIGNIFICANCE LEVELS

Consider a population having distribution F_θ , where θ is unknown, and suppose we want to test a specific hypothesis about θ . We shall denote this hypothesis by H_0 and call it the *null hypothesis*. For example, if F_θ is a normal distribution function with mean θ and variance equal to 1, then two possible null hypotheses about θ are

$$(a) H_0 : \theta = 1$$

$$(b) H_0 : \theta \leq 1$$

Thus the first of these hypotheses states that the population is normal with mean 1 and variance 1, whereas the second states that it is normal with variance 1 and a mean less than or equal to 1. Note that the null hypothesis in (a), when true, completely specifies the population distribution, whereas the null hypothesis in (b) does not. A hypothesis that, when true, completely specifies the population distribution is called a *simple* hypothesis; one that does not is called a *composite* hypothesis.

Suppose now that in order to test a specific null hypothesis H_0 , a population sample of size n — say $X_1, \dots, X_n$ — is to be observed. Based on these n values, we must decide whether or not to accept H_0 . A test for H_0 can be specified by defining a region C in n -dimensional space with the proviso that the hypothesis is to be rejected if the random sample $X_1, \dots, X_n$ turns out to lie in C and accepted otherwise. The region C is called the *critical region*. In other words, the statistical test determined by the critical region C is the one that

$$\text{accepts } H_0 \quad \text{if } (X_1, X_2, \dots, X_n) \notin C$$

and

$$\text{rejects } H_0 \quad \text{if } (X_1, \dots, X_n) \in C$$

For instance, a common test of the hypothesis that θ , the mean of a normal population with variance 1, is equal to 1 has a critical region given by

$$C = \{(X_1, \dots, X_n) : |\bar{X} - 1| > 1.96/\sqrt{n}\} \quad (8.2.1)$$

Thus, this test calls for rejection of the null hypothesis that $\theta = 1$ when the sample average differs from 1 by more than 1.96 divided by the square root of the sample size.

It is important to note when developing a procedure for testing a given null hypothesis H_0 that, in any test, two different types of errors can result. The first of these, called a *type I error*, is said to result if the test incorrectly calls for rejecting H_0 when it is indeed correct. The second, called a *type II error*, results if the test calls for accepting H_0 when it is false.

Now, as was previously mentioned, the objective of a statistical test of H_0 is not to explicitly determine whether or not H_0 is true but rather to determine if its validity is consistent with the resultant data. Hence, with this objective it seems reasonable that H_0 should only be rejected if the resultant data are very unlikely when H_0 is true. The classical way of accomplishing this is to specify a value α and then require the test to have the property that whenever H_0 is true its probability of being rejected is never greater than α . The value α , called the *level of significance of the test*, is usually set in advance, with commonly chosen values being $\alpha = .1, .05, .005$. In other words, the classical approach to testing H_0 is to fix a significance level α and then require that the test have the property that the probability of a type I error occurring can never be greater than α .

Suppose now that we are interested in testing a certain hypothesis concerning θ , an unknown parameter of the population. Specifically, for a given set of parameter values w , suppose we are interested in testing

$$H_0 : \theta \in w$$

A common approach to developing a test of H_0 , say at level of significance α , is to start by determining a point estimator of θ — say $d(\mathbf{X})$. The hypothesis is then rejected if $d(\mathbf{X})$ is “far away” from the region w . However, to determine how “far away” it need be to justify rejection of H_0 , we need to determine the probability distribution of $d(\mathbf{X})$ when H_0 is true since this will usually enable us to determine the appropriate critical region so as to make the test have the required significance level α . For example, the test of the hypothesis that the mean of a normal $(\theta, 1)$ population is equal to 1, given by Equation 8.2.1, calls for rejection when the point estimate of θ — that is, the sample average — is farther than $1.96/\sqrt{n}$ away from 1. As we will see in the next section, the value $1.96/\sqrt{n}$ was chosen to meet a level of significance of $\alpha = .05$.

8.3 TESTS CONCERNING THE MEAN OF A NORMAL POPULATION

8.3.1 CASE OF KNOWN VARIANCE

Suppose that $X_1, \dots, X_n$ is a sample of size n from a normal distribution having an unknown mean μ and a known variance σ^2 and suppose we are interested in testing the null hypothesis

$$H_0 : \mu = \mu_0$$

against the alternative hypothesis

$$H_1 : \mu \neq \mu_0$$

where μ_0 is some specified constant.

Since $\bar{X} = \sum_{i=1}^n X_i/n$ is a natural point estimator of μ , it seems reasonable to accept H_0 if $\bar{X}$ is not too far from μ_0 . That is, the critical region of the test would be of the form

$$C = \{X_1, \dots, X_n : |\bar{X} - \mu_0| > c\} \quad (8.3.1)$$

for some suitably chosen value c .

If we desire that the test has significance level α , then we must determine the critical value c in Equation 8.3.1 that will make the type I error equal to α . That is, c must be such that

$$P_{\mu_0}\{|\bar{X} - \mu_0| > c\} = \alpha \quad (8.3.2)$$

where we write P_{μ_0} to mean that the preceding probability is to be computed under the assumption that $\mu = \mu_0$. However, when $\mu = \mu_0$, $\bar{X}$ will be normally distributed with mean μ_0 and variance σ^2/n and so Z , defined by

$$Z \equiv \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

will have a standard normal distribution. Now Equation 8.3.2 is equivalent to

$$P\left\{|Z| > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

or, equivalently,

$$2P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

where Z is a standard normal random variable. However, we know that

$$P\{Z > z_{\alpha/2}\} = \alpha/2$$

and so

$$\frac{c\sqrt{n}}{\sigma} = z_{\alpha/2}$$

or

$$c = \frac{z_{\alpha/2}\sigma}{\sqrt{n}}$$

Thus, the significance level α test is to reject H_0 if $|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$ and accept otherwise; or, equivalently, to

$$\begin{array}{lll} \text{reject } H_0 & \text{if } \frac{\sqrt{n}}{\sigma}|\bar{X} - \mu_0| > z_{\alpha/2} \\ \text{accept } H_0 & \text{if } \frac{\sqrt{n}}{\sigma}|\bar{X} - \mu_0| \leq z_{\alpha/2} \end{array} \quad (8.3.3)$$

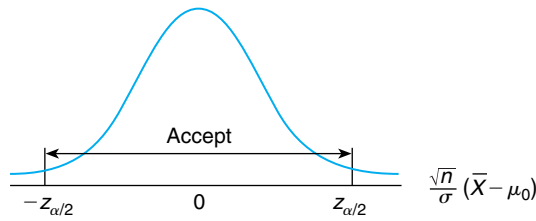


FIGURE 8.1

This can be pictorially represented as shown in Figure 8.1, where we have superimposed the standard normal density function [which is the density of the test statistic $\sqrt{n}(\bar{X} - \mu_0)/\sigma$ when H_0 is true].

EXAMPLE 8.3a It is known that if a signal of value μ is sent from location A, then the value received at location B is normally distributed with mean μ and standard deviation 2. That is, the random noise added to the signal is an $N(0, 4)$ random variable. There is reason for the people at location B to suspect that the signal value $\mu = 8$ will be sent today. Test this hypothesis if the same signal value is independently sent five times and the average value received at location B is $\bar{X} = 9.5$.

SOLUTION Suppose we are testing at the 5 percent level of significance. To begin, we compute the test statistic

$$\frac{\sqrt{n}}{\sigma} |\bar{X} - \mu_0| = \frac{\sqrt{5}}{2} (1.5) = 1.68$$

Since this value is less than $z_{.025} = 1.96$, the hypothesis is accepted. In other words, the data are not inconsistent with the null hypothesis in the sense that a sample average as far from the value 8 as observed would be expected, when the true mean is 8, over 5 percent of the time. Note, however, that if a less stringent significance level were chosen — say $\alpha = .1$ — then the null hypothesis would have been rejected. This follows since $z_{.05} = 1.645$, which is less than 1.68. Hence, if we would have chosen a test that had a 10 percent chance of rejecting H_0 when H_0 was true, then the null hypothesis would have been rejected.

The “correct” level of significance to use in a given situation depends on the individual circumstances involved in that situation. For instance, if rejecting a null hypothesis H_0 would result in large costs that would thus be lost if H_0 were indeed true, then we might elect to be quite conservative and so choose a significance level of .05 or .01. Also, if we initially feel strongly that H_0 was correct, then we would require very stringent data evidence to the contrary for us to reject H_0 . (That is, we would set a very low significance level in this situation.) ■

The test given by Equation 8.3.3 can be described as follows: For any observed value of the test statistic $\sqrt{n}|\bar{X} - \mu_0|/\sigma$, call it v , the test calls for rejection of the null hypothesis if the probability that the test statistic would be as large as v when H_0 is true is less than or equal to the significance level α . From this, it follows that we can determine whether or not to accept the null hypothesis by computing, first, the value of the test statistic and, second, the probability that a standard normal would (in absolute value) exceed that quantity. This probability — called the p -value of the test — gives the critical significance level in the sense that H_0 will be accepted if the significance level α is less than the p -value and rejected if it is greater than or equal.

In practice, the significance level is often not set in advance but rather the data are looked at to determine the resultant p -value. Sometimes, this critical significance level is clearly much larger than any we would want to use, and so the null hypothesis can be readily accepted. At other times the p -value is so small that it is clear that the hypothesis should be rejected.

EXAMPLE 8.3b In Example 8.3a, suppose that the average of the 5 values received is $\bar{X} = 8.5$. In this case,

$$\frac{\sqrt{n}}{\sigma}|\bar{X} - \mu_0| = \frac{\sqrt{5}}{4} = .559$$

Since

$$\begin{aligned} P\{|Z| > .559\} &= 2P\{Z > .559\} \\ &= 2 \times .288 = .576 \end{aligned}$$

it follows that the p -value is .576 and thus the null hypothesis H_0 that the signal sent has value 8 would be accepted at any significance level $\alpha < .576$. Since we would clearly never want to test a null hypothesis using a significance level as large as .576, H_0 would be accepted.

On the other hand, if the average of the data values were 11.5, then the p -value of the test that the mean is equal to 8 would be

$$\begin{aligned} P\{|Z| > 1.75\sqrt{5}\} &= P\{|Z| > 3.913\} \\ &\approx .00005 \end{aligned}$$

For such a small p -value, the hypothesis that the value 8 was sent is rejected. ■

We have not yet talked about the probability of a type II error — that is, the probability of accepting the null hypothesis when the true mean μ is unequal to μ_0 . This probability

will depend on the value of μ , and so let us define $\beta(\mu)$ by

$$\begin{aligned}\beta(\mu) &= P_\mu\{\text{acceptance of } H_0\} \\ &= P_\mu\left\{\left|\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}\right| \leq z_{\alpha/2}\right\} \\ &= P_\mu\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\}\end{aligned}$$

The function $\beta(\mu)$ is called the *operating characteristic* (or OC) *curve* and represents the probability that H_0 will be accepted when the true mean is μ .

To compute this probability, we use the fact that $\bar{X}$ is normal with mean μ and variance σ^2/n and so

$$Z \equiv \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

Hence,

$$\begin{aligned}\beta(\mu) &= P_\mu\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\} \\ &= P_\mu\left\{-z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq \frac{\bar{X} - \mu_0 - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}}\right\} \\ &= P_\mu\left\{-z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq Z - \frac{\mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}}\right\} \\ &= P\left\{\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq Z \leq \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right\} \\ &= \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)\end{aligned}\tag{8.3.4}$$

where Φ is the standard normal distribution function.

For a fixed significance level α , the OC curve given by Equation 8.3.4 is symmetric about μ_0 and indeed will depend on μ only through $\sqrt{n}|\mu - \mu_0|/\sigma$. This curve with the abscissa changed from μ to $d = \sqrt{n}|\mu - \mu_0|/\sigma$ is presented in Figure 8.2 when $\alpha = .05$.

EXAMPLE 8.3c For the problem presented in Example 8.3a, let us determine the probability of accepting the null hypothesis that $\mu = 8$ when the actual value sent is 10. To do so, we compute

$$\frac{\sqrt{n}}{\sigma}(\mu_0 - \mu) = -\frac{\sqrt{5}}{2} \times 2 = -\sqrt{5}$$

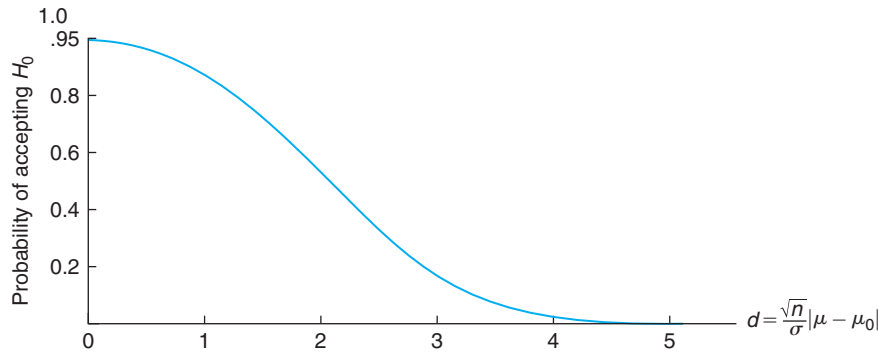


FIGURE 8.2 The OC curve for the two-sided normal test for significance level $\alpha = .05$.

As $z_{.025} = 1.96$, the desired probability is, from Equation 8.3.4,

$$\begin{aligned}
 & \Phi(-\sqrt{5} + 1.96) - \Phi(-\sqrt{5} - 1.96) \\
 &= 1 - \Phi(\sqrt{5} - 1.96) - [1 - \Phi(\sqrt{5} + 1.96)] \\
 &= \Phi(4.196) - \Phi(.276) \\
 &= .392 \quad \blacksquare
 \end{aligned}$$

REMARK

The function $1 - \beta(\mu)$ is called the *power-function* of the test. Thus, for a given value μ , the power of the test is equal to the probability of rejection when μ is the true value. $\blacksquare$

The operating characteristic function is useful in determining how large the random sample need be to meet certain specifications concerning type II errors. For instance, suppose that we desire to determine the sample size n necessary to ensure that the probability of accepting $H_0 : \mu = \mu_0$ when the true mean is actually μ_1 is approximately β . That is, we want n to be such that

$$\beta(\mu_1) \approx \beta$$

But from Equation 8.3.4, this is equivalent to

$$\Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} + z_{\alpha/2}\right) - \Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2}\right) \approx \beta \quad (8.3.5)$$

Although the foregoing cannot be analytically solved for n , a solution can be obtained by using the standard normal distribution table. In addition, an approximation for n can be derived from Equation 8.3.5 as follows. To start, suppose that $\mu_1 > \mu_0$. Then, because this implies that

$$\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2} \leq -z_{\alpha/2}$$

it follows, since Φ is an increasing function, that

$$\Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2}\right) \leq \Phi(-z_{\alpha/2}) = P\{Z \leq -z_{\alpha/2}\} = P\{Z \geq z_{\alpha/2}\} = \alpha/2$$

Hence, we can take

$$\Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2}\right) \approx 0$$

and so from Equation 8.3.5

$$\beta \approx \Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} + z_{\alpha/2}\right) \quad (8.3.6)$$

or, since

$$\beta = P\{Z > z_\beta\} = P\{Z < -z_\beta\} = \Phi(-z_\beta)$$

we obtain from Equation 8.3.6 that

$$-z_\beta \approx (\mu_0 - \mu_1) \frac{\sqrt{n}}{\sigma} + z_{\alpha/2}$$

or

$$n \approx \frac{(z_{\alpha/2} + z_\beta)^2 \sigma^2}{(\mu_1 - \mu_0)^2} \quad (8.3.7)$$

In fact, the same approximation would result when $\mu_1 < \mu_0$ (the details are left as an exercise) and so Equation 8.3.7 is in all cases a reasonable approximation to the sample size necessary to ensure that the type II error at the value $\mu = \mu_1$ is approximately equal to β .

EXAMPLE 8.3d For the problem of Example 8.3a, how many signals need be sent so that the .05 level test of $H_0 : \mu = 8$ has at least a 75 percent probability of rejection when $\mu = 9.2$?

SOLUTION Since $z_{.025} = 1.96$, $z_{.25} = .67$, the approximation 8.3.7 yields

$$n \approx \frac{(1.96 + .67)^2}{(1.2)^2} 4 = 19.21$$

Hence a sample of size 20 is needed. From Equation 8.3.4, we see that with $n = 20$

$$\begin{aligned} \beta(9.2) &= \Phi\left(-\frac{1.2\sqrt{20}}{2} + 1.96\right) - \Phi\left(-\frac{1.2\sqrt{20}}{2} - 1.96\right) \\ &= \Phi(-.723) - \Phi(-4.643) \end{aligned}$$

$$\begin{aligned} &\approx 1 - \Phi(.723) \\ &\approx .235 \end{aligned}$$

Therefore, if the message is sent 20 times, then there is a 76.5 percent chance that the null hypothesis $\mu = 8$ will be rejected when the true mean is 9.2. ■

8.3.1.1 ONE-SIDED TESTS

In testing the null hypothesis that $\mu = \mu_0$, we have chosen a test that calls for rejection when $\bar{X}$ is far from μ_0 . That is, a very small value of $\bar{X}$ or a very large value appears to make it unlikely that μ (which $\bar{X}$ is estimating) could equal μ_0 . However, what happens when the only alternative to μ being equal to μ_0 is for μ to be greater than μ_0 ? That is, what happens when the alternative hypothesis to $H_0 : \mu = \mu_0$ is $H_1 : \mu > \mu_0$? Clearly, in this latter case we would not want to reject H_0 when $\bar{X}$ is small (since a small $\bar{X}$ is more likely when H_0 is true than when H_1 is true). Thus, in testing

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu > \mu_0 \quad (8.3.8)$$

we should reject H_0 when $\bar{X}$, the point estimate of μ_0 , is much greater than μ_0 . That is, the critical region should be of the following form:

$$C = \{(X_1, \dots, X_n) : \bar{X} - \mu_0 > c\}$$

Since the probability of rejection should equal α when H_0 is true (that is, when $\mu = \mu_0$), we require that c be such that

$$P_{\mu_0}\{\bar{X} - \mu_0 > c\} = \alpha \quad (8.3.9)$$

But since

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

has a standard normal distribution when H_0 is true, Equation 8.3.9 is equivalent to

$$P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

when Z is a standard normal. But since

$$P\{Z > z_\alpha\} = \alpha$$

we see that

$$c = \frac{z_\alpha \sigma}{\sqrt{n}}$$

Hence, the test of the hypothesis 8.3.8 is to reject H_0 if $\bar{X} - \mu_0 > z_\alpha \sigma / \sqrt{n}$, and accept otherwise; or, equivalently, to

$$\begin{aligned} \text{accept } H_0 & \quad \text{if } \frac{\sqrt{n}}{\sigma}(\bar{X} - \mu_0) \leq z_\alpha \\ \text{reject } H_0 & \quad \text{if } \frac{\sqrt{n}}{\sigma}(\bar{X} - \mu_0) > z_\alpha \end{aligned} \quad (8.3.10)$$

This is called a *one-sided* critical region (since it calls for rejection only when $\bar{X}$ is large). Correspondingly, the hypothesis testing problem

$$\begin{aligned} H_0 : \mu &= \mu_0 \\ H_1 : \mu &> \mu_0 \end{aligned}$$

is called a one-sided testing problem (in contrast to the *two-sided* problem that results when the alternative hypothesis is $H_1 : \mu \neq \mu_0$).

To compute the p -value in the one-sided test, Equation 8.3.10, we first use the data to determine the value of the statistic $\sqrt{n}(\bar{X} - \mu_0)/\sigma$. The p -value is then equal to the probability that a standard normal would be at least as large as this value.

EXAMPLE 8.3e Suppose in Example 8.3a that we know in advance that the signal value is at least as large as 8. What can be concluded in this case?

SOLUTION To see if the data are consistent with the hypothesis that the mean is 8, we test

$$H_0 : \mu = 8$$

against the one-sided alternative

$$H_1 : \mu > 8$$

The value of the test statistic is $\sqrt{n}(\bar{X} - \mu_0)/\sigma = \sqrt{5}(9.5 - 8)/2 = 1.68$, and the p -value is the probability that a standard normal would exceed 1.68, namely,

$$p\text{-value} = 1 - \Phi(1.68) = .0465$$

the test would call for rejection at all significance levels greater than or equal to .0465, it would, for instance, reject the null hypothesis at the $\alpha = .05$ level of significance. ■

The operating characteristic function of the one-sided test, Equation 8.3.10,

$$\beta(\mu) = P_\mu\{\text{accepting } H_0\}$$

can be obtained as follows:

$$\begin{aligned}\beta(\mu) &= P_\mu \left\{ \bar{X} \leq \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \right\} \\ &= P \left\{ \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \leq \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha \right\} \\ &= P \left\{ Z \leq \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha \right\}, \quad Z \sim \mathcal{N}(0, 1)\end{aligned}$$

where the last equation follows since $\sqrt{n}(\bar{X} - \mu)/\sigma$ has a standard normal distribution. Hence we can write

$$\beta(\mu) = \Phi \left(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha \right)$$

Since Φ , being a distribution function, is increasing in its argument, it follows that $\beta(\mu)$ decreases in μ , which is intuitively pleasing since it certainly seems reasonable that the larger the true mean μ , the less likely it should be to conclude that $\mu \leq \mu_0$. Also since $\Phi(z_\alpha) = 1 - \alpha$, it follows that

$$\beta(\mu_0) = 1 - \alpha$$

The test given by Equation 8.3.10, which was designed to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$, can also be used to test, at level of significance α , the one-sided hypothesis

$$H_0 : \mu \leq \mu_0$$

versus

$$H_1 : \mu > \mu_0$$

To verify that it remains a level α test, we need to show that the probability of rejection is never greater than α when H_0 is true. That is, we must verify that

$$1 - \beta(\mu) \leq \alpha \quad \text{for all } \mu \leq \mu_0$$

or

$$\beta(\mu) \geq 1 - \alpha \quad \text{for all } \mu \leq \mu_0$$

But it has previously been shown that for the test given by Equation 8.3.10, $\beta(\mu)$ decreases in μ and $\beta(\mu_0) = 1 - \alpha$. This gives that

$$\beta(\mu) \geq \beta(\mu_0) = 1 - \alpha \quad \text{for all } \mu \leq \mu_0$$

which shows that the test given by Equation 8.3.10 remains a level α test for $H_0 : \mu \leq \mu_0$ against the alternative hypothesis $H_1 : \mu > \mu_0$.

REMARK

We can also test the one-sided hypothesis

$$H_0 : \mu = \mu_0 \quad (\text{or } \mu \geq \mu_0) \quad \text{versus} \quad H_1 : \mu < \mu_0$$

at significance level α by

$$\begin{aligned} &\text{accepting } H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\bar{X} - \mu_0) \geq -z_\alpha \\ &\text{rejecting } H_0 \quad \text{otherwise} \end{aligned}$$

This test can alternatively be performed by first computing the value of the test statistic $\sqrt{n}(\bar{X} - \mu_0)/\sigma$. The p -value would then equal the probability that a standard normal would be less than this value, and the hypothesis would be rejected at any significance level greater than or equal to this p -value.

EXAMPLE 8.3f All cigarettes presently on the market have an average nicotine content of at least 1.6 mg per cigarette. A firm that produces cigarettes claims that it has discovered a new way to cure tobacco leaves that will result in the average nicotine content of a cigarette being less than 1.6 mg. To test this claim, a sample of 20 of the firm's cigarettes were analyzed. If it is known that the standard deviation of a cigarette's nicotine content is .8 mg, what conclusions can be drawn, at the 5 percent level of significance, if the average nicotine content of the 20 cigarettes is 1.54?

Note: The above raises the question of how we would know in advance that the standard deviation is .8. One possibility is that the variation in a cigarette's nicotine content is due to variability in the amount of tobacco in each cigarette and not on the method of curing that is used. Hence, the standard deviation can be known from previous experience.

SOLUTION We must first decide on the appropriate null hypothesis. As was previously noted, our approach to testing is not symmetric with respect to the null and the alternative hypotheses since we consider only tests having the property that their probability of rejecting the null hypothesis when it is true will never exceed the significance level α . Thus, whereas rejection of the null hypothesis is a strong statement about the data not being consistent with this hypothesis, an analogous statement cannot be made when the null hypothesis is accepted. Hence, since in the preceding example we would like to endorse the producer's claims only when there is substantial evidence for it, we should take this claim as the alternative hypothesis. That is, we should test

$$H_0 : \mu \geq 1.6 \quad \text{versus} \quad H_1 : \mu < 1.6$$

Now, the value of the test statistic is

$$\sqrt{n}(\bar{X} - \mu_0)/\sigma = \sqrt{20}(1.54 - 1.6)/.8 = -.336$$

and so the p -value is given by

$$\begin{aligned} p\text{-value} &= P\{Z < -.336\}, \quad Z \sim N(0, 1) \\ &= .368 \end{aligned}$$

Since this value is greater than .05, the foregoing data do not enable us to reject, at the .05 percent level of significance, the hypothesis that the mean nicotine content exceeds 1.6 mg. In other words, the evidence, although supporting the cigarette producer's claim, is not strong enough to prove that claim. ■

REMARKS

(a) There is a direct analogy between confidence interval estimation and hypothesis testing. For instance, for a normal population having mean μ and known variance σ^2 , we have shown in Section 7.3 that a $100(1 - \alpha)$ percent confidence interval for μ is given by

$$\mu \in \left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

where $\bar{x}$ is the observed sample mean. More formally, the preceding confidence interval statement is equivalent to

$$P \left\{ \mu \in \left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) \right\} = 1 - \alpha$$

Hence, if $\mu = \mu_0$, then the probability that μ_0 will fall in the interval

$$\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

is $1 - \alpha$, implying that a significance level α test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ is to reject H_0 when

$$\mu_0 \notin \left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

Similarly, since a $100(1 - \alpha)$ percent one-sided confidence interval for μ is given by

$$\mu \in \left(\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right)$$

it follows that an α -level significance test of $H_0 : \mu \leq \mu_0$ versus $H_1 : \mu > \mu_0$ is to reject H_0 when $\mu_0 \notin (\bar{X} - z_{\alpha} \sigma / \sqrt{n}, \infty)$ — that is, when $\mu_0 < \bar{X} - z_{\alpha} \sigma / \sqrt{n}$.

TABLE 8.1 $X_1, \dots, X_n$ Is a Sample from a $\mathcal{N}(\mu, \sigma^2)$ Population σ^2 Is Known, $\bar{X} = \sum_{i=1}^n X_i/n$

H_0	H_1	Test Statistic TS	Significance Level α Test	p -Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\bar{X} - \mu_0)/\sigma$	Reject if $ TS > z_{\alpha/2}$	$2P\{Z \geq t \}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\bar{X} - \mu_0)/\sigma$	Reject if $TS > z_{\alpha}$	$P\{Z \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\bar{X} - \mu_0)/\sigma$	Reject if $TS < -z_{\alpha}$	$P\{Z \leq t\}$

 Z is a standard normal random variable.

(b) A Remark on Robustness A test that performs well even when the underlying assumptions on which it is based are violated is said to be *robust*. For instance, the tests of Sections 8.3.1 and 8.3.1.1 were derived under the assumption that the underlying population distribution is normal with known variance σ^2 . However, in deriving these tests, this assumption was used only to conclude that $\bar{X}$ also has a normal distribution. But, by the central limit theorem, it follows that for a reasonably large sample size, $\bar{X}$ will approximately have a normal distribution no matter what the underlying distribution. Thus we can conclude that these tests will be relatively robust for any population distribution with variance σ^2 .

Table 8.1 summarizes the tests of this subsection.

8.3.2 CASE OF UNKNOWN VARIANCE: THE t -TEST

Up to now we have supposed that the only unknown parameter of the normal population distribution is its mean. However, the more common situation is one where the mean μ and variance σ^2 are both unknown. Let us suppose this to be the case and again consider a test of the hypothesis that the mean is equal to some specified value μ_0 . That is, consider a test of

$$H_0 : \mu = \mu_0$$

versus the alternative

$$H_1 : \mu \neq \mu_0$$

It should be noted that the null hypothesis is not a simple hypothesis since it does not specify the value of σ^2 .

As before, it seems reasonable to reject H_0 when the sample mean $\bar{X}$ is far from μ_0 . However, how far away it need be to justify rejection will depend on the variance σ^2 . Recall that when the value of σ^2 was known, the test called for rejecting H_0 when $|\bar{X} - \mu_0|$ exceeded $z_{\alpha/2}\sigma/\sqrt{n}$ or, equivalently, when

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2}$$

Now when σ^2 is no longer known, it seems reasonable to estimate it by

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

and then to reject H_0 when

$$\left| \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \right|$$

is large.

To determine how large a value of the statistic

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right|$$

to require for rejection, in order that the resulting test have significance level α , we must determine the probability distribution of this statistic when H_0 is true. However, as shown in Section 6.5, the statistic T , defined by

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S}$$

has, when $\mu = \mu_0$, a t -distribution with $n-1$ degrees of freedom. Hence,

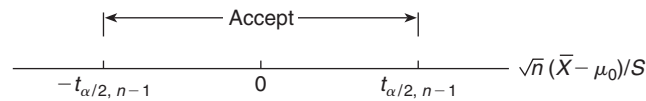
$$P_{\mu_0} \left\{ -t_{\alpha/2, n-1} \leq \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \leq t_{\alpha/2, n-1} \right\} = 1 - \alpha \quad (8.3.11)$$

where $t_{\alpha/2, n-1}$ is the 100 $\alpha/2$ upper percentile value of the t -distribution with $n-1$ degrees of freedom. (That is, $P\{T_{n-1} \geq t_{\alpha/2, n-1}\} = P\{T_{n-1} \leq -t_{\alpha/2, n-1}\} = \alpha/2$ when T_{n-1} has a t -distribution with $n-1$ degrees of freedom.) From Equation 8.3.11 we see that the appropriate significance level α test of

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

is, when σ^2 is unknown, to

$$\begin{aligned} \text{accept } H_0 & \quad \text{if} \quad \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right| \leq t_{\alpha/2, n-1} \\ \text{reject } H_0 & \quad \text{if} \quad \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right| > t_{\alpha/2, n-1} \end{aligned} \quad (8.3.12)$$

FIGURE 8.3 The two-sided t -test.

The test defined by Equation 8.3.12 is called a *two-sided t -test*. It is pictorially illustrated in Figure 8.3.

If we let t denote the observed value of the test statistic $T = \sqrt{n}(\bar{X} - \mu_0)/S$, then the p -value of the test is the probability that $|T|$ would exceed $|t|$ when H_0 is true. That is, the p -value is the probability that the absolute value of a t -random variable with $n - 1$ degrees of freedom would exceed $|t|$. The test then calls for rejection at all significance levels higher than the p -value and acceptance at all lower significance levels.

Program 8.3.2 computes the value of the test statistic and the corresponding p -value. It can be applied both for one- and two-sided tests. (The one-sided material will be presented shortly.)

EXAMPLE 8.3g Among a clinic's patients having blood cholesterol levels ranging in the medium to high range (at least 220 milliliters per deciliter of serum), volunteers were recruited to test a new drug designed to reduce blood cholesterol. A group of 50 volunteers was given the drug for 1 month and the changes in their blood cholesterol levels were noted. If the average change was a reduction of 14.8 with a sample standard deviation of 6.4, what conclusions can be drawn?

SOLUTION Let us start by testing the hypothesis that the change could be due solely to chance — that is, that the 50 changes constitute a normal sample with mean 0. Because the value of the t -statistic used to test the hypothesis that a normal mean is equal to 0 is

$$T = \sqrt{n} \bar{X}/S = \sqrt{50} 14.8/6.4 = 16.352$$

is clear that we should reject the hypothesis that the changes were solely due to chance. Unfortunately, however, we are not justified at this point in concluding that the changes were due to the specific drug used and not to some other possibility. For instance, it is well known that any medication received by a patient (whether or not this medication is directly relevant to the patient's suffering) often leads to an improvement in the patient's condition — the so-called placebo effect. Also, another possibility that may need to be taken into account would be the weather conditions during the month of testing, for it is certainly conceivable that this affects blood cholesterol level. Indeed, it must be concluded that the foregoing was a very poorly designed experiment, for in order to test whether a specific treatment has an effect on a disease that may be affected by many things, we should try to design the experiment so as to neutralize all other possible causes. The accepted approach for accomplishing this is to divide the volunteers at random into two

groups — one group to receive the drug and the other to receive a placebo (that is, a tablet that looks and tastes like the actual drug but has no physiological effect). The volunteers should not be told whether they are in the actual or control group, and indeed it is best if even the clinicians do not have this information (the so-called double-blind test) so as not to allow their own biases to play a role. Since the two groups are chosen at random from among the volunteers, we can now hope that on average all factors affecting the two groups will be the same except that one received the actual drug and the other a placebo. Hence, any difference in performance between the groups can be attributed to the drug. ■

EXAMPLE 8.3h A public health official claims that the mean home water use is 350 gallons a day. To verify this claim, a study of 20 randomly selected homes was instigated with the result that the average daily water uses of these 20 homes were as follows:

340	344	362	375
356	386	354	364
332	402	340	355
362	322	372	324
318	360	338	370

Do the data contradict the official's claim?

SOLUTION To determine if the data contradict the official's claim, we need to test

$$H_0 : \mu = 350 \quad \text{versus} \quad H_1 : \mu \neq 350$$

This can be accomplished by running Program 8.3.2 or, if it is inconvenient to utilize, by noting first that the sample mean and sample standard deviation of the preceding data set are

$$\bar{X} = 353.8, \quad S = 21.8478$$

Thus, the value of the test statistic is

$$T = \frac{\sqrt{20}(3.8)}{21.8478} = .7778$$

Because this is less than $t_{.05,19} = 1.730$, the null hypothesis is accepted at the 10 percent level of significance. Indeed, the p -value of the test data is

$$p\text{-value} = P\{|T_{19}| > .7778\} = 2P\{T_{19} > .7778\} = .4462$$

indicating that the null hypothesis would be accepted at any reasonable significance level, and thus that the data are not inconsistent with the claim of the health official. ■

We can use a one-sided t -test to test the hypothesis

$$H_0 : \mu = \mu_0 \quad (\text{or } H_0 : \mu \leq \mu_0)$$

against the one-sided alternative

$$H_1 : \mu > \mu_0$$

The significance level α test is to

$$\begin{aligned} \text{accept } H_0 & \text{ if } \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \leq t_{\alpha, n-1} \\ \text{reject } H_0 & \text{ if } \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > t_{\alpha, n-1} \end{aligned} \quad (8.3.13)$$

If $\sqrt{n}(\bar{X} - \mu_0)/S = v$, then the p -value of the test is the probability that a t -random variable with $n - 1$ degrees of freedom would be at least as large as v .

The significance level α test of

$$H_0 : \mu = \mu_0 \quad (\text{or } H_0 : \mu \geq \mu_0)$$

versus the alternative

$$H_1 : \mu < \mu_0$$

is to

$$\begin{aligned} \text{accept } H_0 & \text{ if } \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \geq -t_{\alpha, n-1} \\ \text{reject } H_0 & \text{ if } \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} < -t_{\alpha, n-1} \end{aligned}$$

The p -value of this test is the probability that a t -random variable with $n - 1$ degrees of freedom would be less than or equal to the observed value of $\sqrt{n}(\bar{X} - \mu_0)/S$.

EXAMPLE 8.3i The manufacturer of a new fiberglass tire claims that its average life will be at least 40,000 miles. To verify this claim a sample of 12 tires is tested, with their lifetimes (in 1,000s of miles) being as follows:

Tire	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>	<u>10</u>	<u>11</u>	<u>12</u>
Life	36.1	40.2	33.8	38.5	42	35.8	37	41	36.8	37.2	33	36

Test the manufacturer's claim at the 5 percent level of significance.

SOLUTION To determine whether the foregoing data are consistent with the hypothesis that the mean life is at least 40,000 miles, we will test

$$H_0 : \mu \geq 40,000 \quad \text{versus} \quad H_1 : \mu < 40,000$$

A computation gives that

$$\bar{X} = 37.2833, \quad S = 2.7319$$

and so the value of the test statistic is

$$T = \frac{\sqrt{12}(37.2833 - 40)}{2.7319} = -3.4448$$

Since this is less than $-t_{.05,11} = -1.796$, the null hypothesis is rejected at the 5 percent level of significance. Indeed, the p -value of the test data is

$$p\text{-value} = P\{T_{11} < -3.4448\} = P\{T_{11} > 3.4448\} = .0028$$

indicating that the manufacturer’s claim would be rejected at any significance level greater than .003. ■

The preceding could also have been obtained by using Program 8.3.2, as illustrated in Figure 8.4.

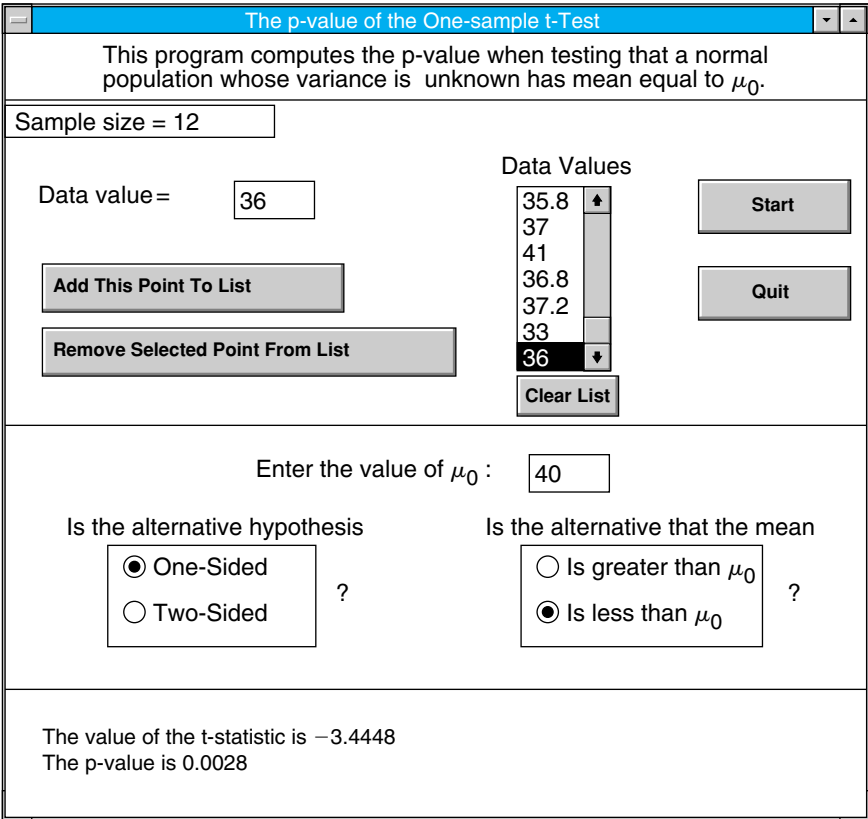


FIGURE 8.4

EXAMPLE 8.3j In a single-server queueing system in which customers arrive according to a Poisson process, the long-run average queueing delay per customer depends on the service distribution through its mean and variance. Indeed, if μ is the mean service time, and σ^2 is the variance of a service time, then the average amount of time that a customer spends waiting in queue is given by

$$\frac{\lambda(\mu^2 + \sigma^2)}{2(1 - \lambda\mu)}$$

provided that $\lambda\mu < 1$, where λ is the arrival rate. (The average delay is infinite if $\lambda\mu \geq 1$.) As can be seen by this formula, the average delay is quite large when μ is only slightly smaller than $1/\lambda$, where, since λ is the arrival *rate*, $1/\lambda$ is the average time between arrivals.

Suppose that the owner of a service station will hire a second server if it can be shown that the average service time exceeds 8 minutes. The following data give the service times (in minutes) of 28 customers of this queueing system. Do they indicate that the mean service time is greater than 8 minutes?

8.6, 9.4, 5.0, 4.4, 3.7, 11.4, 10.0, 7.6, 14.4, 12.2, 11.0, 14.4, 9.3, 10.5,
10.3, 7.7, 8.3, 6.4, 9.2, 5.7, 7.9, 9.4, 9.0, 13.3, 11.6, 10.0, 9.5, 6.6

SOLUTION Let us use the preceding data to test the null hypothesis that the mean service time is less than or equal to 8 minutes. A small p -value will then be strong evidence that the mean service time is greater than 8 minutes. Running Program 8.3.2 on these data shows that the value of the test statistic is 2.257, with a resulting p -value of .016. Such a small p -value is certainly strong evidence that the mean service time exceeds 8 minutes. ■

Table 8.2 summarizes the tests of this subsection.

TABLE 8.2 $X_1, \dots, X_n$ Is a Sample from a $\mathcal{N}(\mu, \sigma^2)$ Population

σ^2 Is Unknown, $\bar{X} = \sum_{i=1}^n X_i / n$, $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1)$

H_0	H_1	Test Statistic TS	Significance Level α Test	p -Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\bar{X} - \mu_0)/S$	Reject if $ TS > t_{\alpha/2, n-1}$	$2P\{T_{n-1} \geq t \}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\bar{X} - \mu_0)/S$	Reject if $TS > t_{\alpha, n-1}$	$P\{T_{n-1} \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\bar{X} - \mu_0)/S$	Reject if $TS < -t_{\alpha, n-1}$	$P\{T_{n-1} \leq t\}$

T_{n-1} is a t -random variable with $n - 1$ degrees of freedom: $P\{T_{n-1} > t_{\alpha, n-1}\} = \alpha$.

8.4 TESTING THE EQUALITY OF MEANS OF TWO NORMAL POPULATIONS

A common situation faced by a practicing engineer is one in which she must decide whether two different approaches lead to the same solution. Often such a situation can be modeled as a test of the hypothesis that two normal populations have the same mean value.

8.4.1 CASE OF KNOWN VARIANCES

Suppose that $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ are independent samples from normal populations having unknown means μ_x and μ_y but known variances σ_x^2 and σ_y^2 . Let us consider the problem of testing the hypothesis

$$H_0 : \mu_x = \mu_y$$

versus the alternative

$$H_1 : \mu_x \neq \mu_y$$

Since $\bar{X}$ is an estimate of μ_x and $\bar{Y}$ of μ_y , it follows that $\bar{X} - \bar{Y}$ can be used to estimate $\mu_x - \mu_y$. Hence, because the null hypothesis can be written as $H_0 : \mu_x - \mu_y = 0$, it seems reasonable to reject it when $\bar{X} - \bar{Y}$ is far from zero. That is, the form of the test should be to

$$\begin{aligned} &\text{reject } H_0 \quad \text{if } |\bar{X} - \bar{Y}| > c \\ &\text{accept } H_0 \quad \text{if } |\bar{X} - \bar{Y}| \leq c \end{aligned} \tag{8.4.1}$$

for some suitably chosen value c .

To determine that value of c that would result in the test in Equations 8.4.1 having a significance level α , we need determine the distribution of $\bar{X} - \bar{Y}$ when H_0 is true. However, as was shown in Section 7.3.2,

$$\bar{X} - \bar{Y} \sim \mathcal{N}\left(\mu_x - \mu_y, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

which implies that

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim \mathcal{N}(0, 1) \tag{8.4.2}$$

Hence, when H_0 is true (and so $\mu_x - \mu_y = 0$), it follows that

$$(\bar{X} - \bar{Y}) / \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

has a standard normal distribution, and thus

$$P_{H_0} \left\{ -z_{\alpha/2} \leq \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \leq z_{\alpha/2} \right\} = 1 - \alpha \quad (8.4.3)$$

From Equation 8.4.3, we obtain that the significance level α test of $H_0 : \mu_x = \mu_y$ versus $H_1 : \mu_x \neq \mu_y$ is

$$\begin{aligned} \text{accept } H_0 & \text{ if } \frac{|\bar{X} - \bar{Y}|}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}} \leq z_{\alpha/2} \\ \text{reject } H_0 & \text{ if } \frac{|\bar{X} - \bar{Y}|}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}} \geq z_{\alpha/2} \end{aligned}$$

Program 8.4.1 will compute the value of the test statistic $(\bar{X} - \bar{Y}) / \sqrt{\sigma_x^2/n + \sigma_y^2/m}$.

EXAMPLE 8.4a Two new methods for producing a tire have been proposed. To ascertain which is superior, a tire manufacturer produces a sample of 10 tires using the first method and a sample of 8 using the second. The first set is to be road tested at location A and the second at location B. It is known from past experience that the lifetime of a tire that is road tested at one of these locations is normally distributed with a mean life due to the tire but with a variance due (for the most part) to the location. Specifically, it is known that the lifetimes of tires tested at location A are normal with standard deviation equal to 4,000 kilometers, whereas those tested at location B are normal with $\sigma = 6,000$ kilometers. If the manufacturer is interested in testing the hypothesis that there is no appreciable difference in the mean life of tires produced by either method, what conclusion should be drawn at the 5 percent level of significance if the resulting data are as given in Table 8.3?

TABLE 8.3 Tire Lives in Units of 100 Kilometers

Tires Tested at A	Tires Tested at B
61.1	62.2
58.2	56.6
62.3	66.4
64	56.2
59.7	57.4
66.2	58.4
57.8	57.6
61.4	65.4
62.2	
63.6	

SOLUTION A simple computation (or the use of Program 8.4.1) shows that the value of the test statistic is .066. For such a small value of the test statistic (which has a standard normal distribution when H_0 is true), it is clear that the null hypothesis is accepted. ■

It follows from Equation 8.4.1 that a test of the hypothesis $H_0 : \mu_x = \mu_y$ (or $H_0 : \mu_x \leq \mu_y$) against the one-sided alternative $H_1 : \mu_x > \mu_y$ would be to

$$\begin{aligned} \text{accept } H_0 & \text{ if } \bar{X} - \bar{Y} \leq z_\alpha \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}} \\ \text{reject } H_0 & \text{ if } \bar{X} - \bar{Y} > z_\alpha \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}} \end{aligned}$$

8.4.2 CASE OF UNKNOWN VARIANCES

Suppose again that $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ are independent samples from normal populations having respective parameters (μ_x, σ_x^2) and (μ_y, σ_y^2) , but now suppose that all four parameters are unknown. We will once again consider a test of

$$H_0 : \mu_x = \mu_y \quad \text{versus} \quad H_1 : \mu_x \neq \mu_y$$

To determine a significance level α test of H_0 we will need to make the additional assumption that the unknown variances σ_x^2 and σ_y^2 are equal. Let σ^2 denote their value — that is,

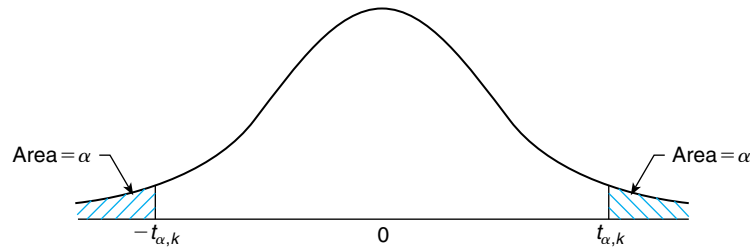
$$\sigma^2 = \sigma_x^2 = \sigma_y^2$$

As before, we would like to reject H_0 when $\bar{X} - \bar{Y}$ is “far” from zero. To determine how far from zero it needs to be, let

$$\begin{aligned} S_x^2 &= \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} \\ S_y^2 &= \frac{\sum_{i=1}^m (Y_i - \bar{Y})^2}{m-1} \end{aligned}$$

denote the sample variances of the two samples. Then, as was shown in Section 7.3.2,

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{S_p^2(1/n + 1/m)}} \sim t_{n+m-2}$$

FIGURE 8.5 Density of a t -random variable with k degrees of freedom.

where S_p^2 , the *pooled* estimator of the common variance σ^2 , is given by

$$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$$

Hence, when H_0 is true, and so $\mu_x - \mu_y = 0$, the statistic

$$T \equiv \frac{\bar{X} - \bar{Y}}{\sqrt{S_p^2(1/n + 1/m)}}$$

has a t -distribution with $n + m - 2$ degrees of freedom. From this, it follows that we can test the hypothesis that $\mu_x = \mu_y$ as follows:

$$\begin{aligned} \text{accept } H_0 & \text{ if } |T| \leq t_{\alpha/2, n+m-2} \\ \text{reject } H_0 & \text{ if } |T| > t_{\alpha/2, n+m-2} \end{aligned}$$

where $t_{\alpha/2, n+m-2}$ is the 100 $\alpha/2$ percentile point of a t -random variable with $n + m - 2$ degrees of freedom (see Figure 8.5).

Alternatively, the test can be run by determining the p -value. If T is observed to equal v , then the resulting p -value of the test of H_0 against H_1 is given by

$$\begin{aligned} p\text{-value} &= P\{|T_{n+m-2}| \geq |v|\} \\ &= 2P\{T_{n+m-2} \geq |v|\} \end{aligned}$$

where T_{n+m-2} is a t -random variable having $n + m - 2$ degrees of freedom.

If we are interested in testing the one-sided hypothesis

$$H_0 : \mu_x \leq \mu_y \quad \text{versus} \quad H_1 : \mu_x > \mu_y$$

then H_0 will be rejected at large values of T . Thus the significance level α test is to

$$\begin{aligned} \text{reject } H_0 & \text{ if } T \geq t_{\alpha, n+m-2} \\ \text{not reject } H_0 & \text{ otherwise} \end{aligned}$$

If the value of the test statistic T is v , then the p -value is given by

$$p\text{-value} = P\{T_{n+m-2} \geq v\}$$

Program 8.4.2 computes both the value of the test statistic and the corresponding p -value.

EXAMPLE 8.4b Twenty-two volunteers at a cold research institute caught a cold after having been exposed to various cold viruses. A random selection of 10 of these volunteers was given tablets containing 1 gram of vitamin C. These tablets were taken four times a day. The control group consisting of the other 12 volunteers was given placebo tablets that looked and tasted exactly the same as the vitamin C tablets. This was continued for each volunteer until a doctor, who did not know if the volunteer was receiving the vitamin C or the placebo tablets, decided that the volunteer was no longer suffering from the cold. The length of time the cold lasted was then recorded.

At the end of this experiment, the following data resulted.

Treated with Vitamin C	Treated with Placebo
5.5	6.5
6.0	6.0
7.0	8.5
6.0	7.0
7.5	6.5
6.0	8.0
7.5	7.5
5.5	6.5
7.0	7.5
6.5	6.0
	8.5
	7.0

Do the data listed prove that taking 4 grams daily of vitamin C reduces the mean length of time a cold lasts? At what level of significance?

SOLUTION To prove the above hypothesis, we would need to reject the null hypothesis in a test of

$$H_0 : \mu_p \leq \mu_c \quad \text{versus} \quad H_1 : \mu_p > \mu_c$$

where μ_c is the mean time a cold lasts when the vitamin C tablets are taken and μ_p is the mean time when the placebo is taken. Assuming that the variance of the length of the cold is the same for the vitamin C patients and the placebo patients, we test the above by running Program 8.4.2. This yields the information shown in Figure 8.6. Thus H_0 would be rejected at the 5 percent level of significance.

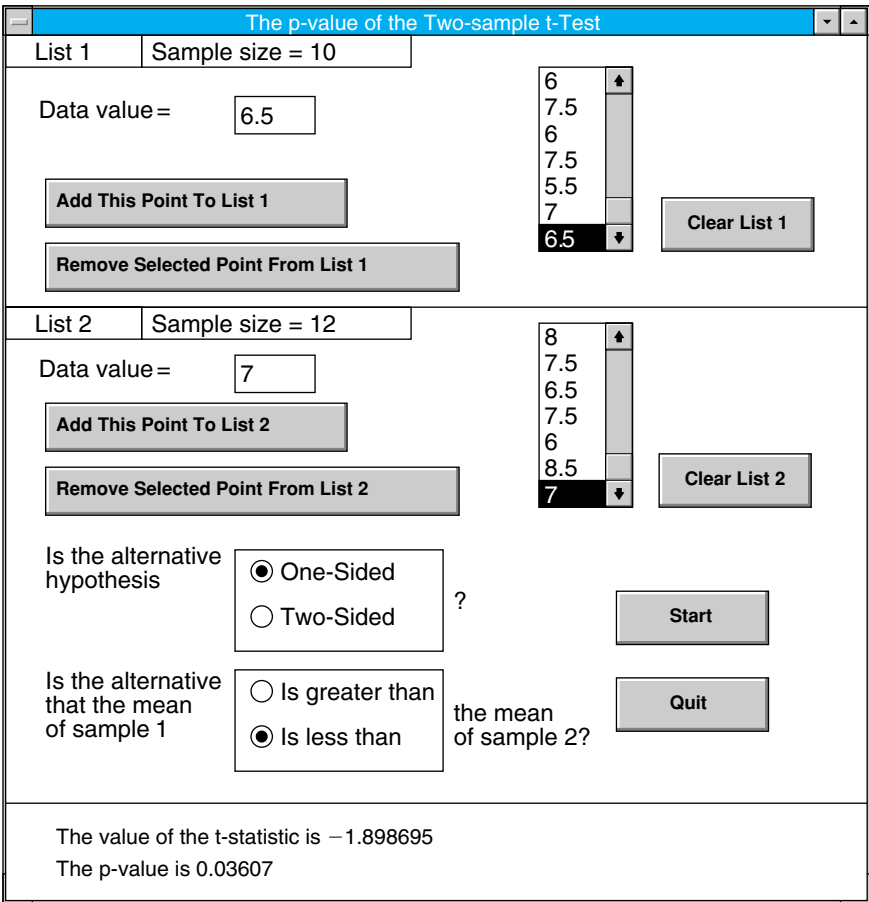


FIGURE 8.6

Of course, if it were not convenient to run Program 8.4.2 then we could have performed the test by first computing the values of the statistics $\bar{X}$, $\bar{Y}$, S_x^2 , S_y^2 , and S_p^2 , where the X sample corresponds to those receiving vitamin C and the Y sample to those receiving a placebo. These computations would give the values

$$\begin{aligned}\bar{X} &= 6.450, & \bar{Y} &= 7.125 \\ S_x^2 &= .581, & S_y^2 &= .778\end{aligned}$$

Therefore,

$$S_p^2 = \frac{9}{20} S_x^2 + \frac{11}{20} S_y^2 = .689$$

and the value of the test statistic is

$$TS = \frac{-.675}{\sqrt{.689(1/10 + 1/12)}} = -1.90$$

Since $t_{.05,20} = 1.725$, the null hypothesis is rejected at the 5 percent level of significance. That is, at the 5 percent level of significance the evidence is significant in establishing that vitamin C reduces the mean time that a cold persists. ■

EXAMPLE 8.4c Reconsider Example 8.4a, but now suppose that the population variances are unknown but equal.

SOLUTION Using Program 8.4.2 yields that the value of the test statistic is 1.028, and the resulting p -value is

$$p\text{-value} = P\{T_{16} > 1.028\} = .3192$$

Thus, the null hypothesis is accepted at any significance level less than .3192. ■

8.4.3 CASE OF UNKNOWN AND UNEQUAL VARIANCES

Let us now suppose that the population variances σ_x^2 and σ_y^2 are not only unknown but also cannot be considered to be equal. In this situation, since S_x^2 is the natural estimator of σ_x^2 and S_y^2 of σ_y^2 , it would seem reasonable to base our test of

$$H_0 : \mu_x = \mu_y \quad \text{versus} \quad H_1 : \mu_x \neq \mu_y$$

on the test statistic

$$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}} \quad (8.4.4)$$

However, the foregoing has a complicated distribution, which, even when H_0 is true, depends on the unknown parameters, and thus cannot be generally employed. The one situation in which we can utilize the statistic of Equation 8.4.4 is when n and m are both large. In such a case, it can be shown that when H_0 is true Equation 8.4.4 will have *approximately* a standard normal distribution. Hence, when n and m are large an *approximate* level α test of $H_0 : \mu_x = \mu_y$ versus $H_1 : \mu_x \neq \mu_y$ is to

$$\begin{array}{ll} \text{accept } H_0 & \text{if } -z_{\alpha/2} \leq \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}} \leq z_{\alpha/2} \\ \text{reject} & \text{otherwise} \end{array}$$

The problem of determining an exact level α test of the hypothesis that the means of two normal populations, having unknown and not necessarily equal variances, are equal is known as the Behrens-Fisher problem. There is no completely satisfactory solution known.

Table 8.4 presents the two-sided tests of this section.

TABLE 8.4 $X_1, \dots, X_n$ Is a Sample from a $\mathcal{N}(\mu_1, \sigma_1^2)$ Population; $Y_1, \dots, Y_m$ Is a Sample from a $\mathcal{N}(\mu_2, \sigma_2^2)$ Population

The Two Population Samples Are Independent to Test $H_0 : \mu_1 = \mu_2$ versus $H_0 : \mu_1 \neq \mu_2$			
Assumption	Test Statistic TS	Significance Level α Test	p -Value if $TS = t$
σ_1, σ_2 known	$\frac{\bar{X} - \bar{Y}}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}}$	Reject if $ TS > z_{\alpha/2}$	$2P\{Z \geq t \}$
$\sigma_1 = \sigma_2$	$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}} \sqrt{1/n + 1/m}}$	Reject if $ TS > t_{\alpha/2, n+m-2}$	$2P\{T_{n+m-2} \geq t \}$
n, m large	$\frac{\bar{X} - \bar{Y}}{\sqrt{S_1^2/n + S_2^2/m}}$	Reject if $ TS > z_{\alpha/2}$	$2P\{Z \geq t \}$

8.4.4 THE PAIRED t -TEST

Suppose we are interested in determining whether the installation of a certain antipollution device will affect a car's mileage. To test this, a collection of n cars that do not have this device are gathered. Each car's mileage per gallon is then determined both before and after the device is installed. How can we test the hypothesis that the antipollution control has no effect on gas consumption?

The data can be described by the n pairs (X_i, Y_i) , $i = 1, \dots, n$, where X_i is the gas consumption of the i th car before installation of the pollution control device, and Y_i of the same car after installation. Because each of the n cars will be inherently different, we cannot treat $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ as being independent samples. For example, if we know that X_1 is large (say, 40 miles per gallon), we would certainly expect that Y_1 would also probably be large. Thus, we cannot employ the earlier methods presented in this section.

One way in which we can test the hypothesis that the antipollution device does not affect gas mileage is to let the data consist of each car's difference in gas mileage. That is, let $W_i = X_i - Y_i$, $i = 1, \dots, n$. Now, if there is no effect from the device, it should follow that the W_i would have mean 0. Hence, we can test the hypothesis of no effect by testing

$$H_0 : \mu_w = 0 \quad \text{versus} \quad H_1 : \mu_w \neq 0$$

where $W_1, \dots, W_n$ are assumed to be a sample from a normal population having unknown mean μ_w and unknown variance σ_w^2 . But the t -test described in Section 8.3.2 shows that this can be tested by

$$\begin{aligned} &\text{accepting } H_0 \quad \text{if} \quad -t_{\alpha/2, n-1} < \sqrt{n} \frac{\overline{W}}{S_w} < t_{\alpha/2, n-1} \\ &\text{rejecting } H_0 \quad \text{otherwise} \end{aligned}$$

EXAMPLE 8.4d An industrial safety program was recently instituted in the computer chip industry. The average weekly loss (averaged over 1 month) in labor-hours due to accidents in 10 similar plants both before and after the program are as follows:

Plant	Before	After	$A - B$
1	30.5	23	-7.5
2	18.5	21	2.5
3	24.5	22	-2.5
4	32	28.5	-3.5
5	16	14.5	-1.5
6	15	15.5	.5
7	23.5	24.5	1
8	25.5	21	-4.5
9	28	23.5	-4.5
10	18	16.5	-1.5

Determine, at the 5 percent level of significance, whether the safety program has been proven to be effective.

SOLUTION To determine this, we will test

$$H_0 : \mu_A - \mu_B \geq 0 \quad \text{versus} \quad H_1 : \mu_A - \mu_B < 0$$

because this will enable us to see whether the null hypothesis that the safety program has not had a beneficial effect is a reasonable possibility. To test this, we run Program 8.3.2, which gives the value of the test statistic as -2.266 , with

$$p\text{-value} = P\{T_q \leq -2.266\} = .025$$

Since the p -value is less than .05, the hypothesis that the safety program has not been effective is rejected and so we can conclude that its effectiveness has been established (at least for any significance level greater than .025). ■

Note that the paired-sample t -test can be used even though the samples are not independent and the population variances are unequal.

8.5 HYPOTHESIS TESTS CONCERNING THE VARIANCE OF A NORMAL POPULATION

Let $X_1, \dots, X_n$ denote a sample from a normal population having unknown mean μ and unknown variance σ^2 , and suppose we desire to test the hypothesis

$$H_0 : \sigma^2 = \sigma_0^2$$

versus the alternative

$$H_1 : \sigma^2 \neq \sigma_0^2$$

for some specified value σ_0^2 .

To obtain a test, recall (as was shown in Section 6.5) that $(n-1)S^2/\sigma^2$ has a chi-square distribution with $n-1$ degrees of freedom. Hence, when H_0 is true

$$\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$$

Because $P\{\chi_{n-1}^2 < \chi_{\alpha/2, n-1}^2\} = 1 - \alpha/2$ and $P\{\chi_{n-1}^2 < \chi_{1-\alpha/2, n-1}^2\} = \alpha/2$, it follows that

$$P_{H_0} \left\{ \chi_{1-\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi_{\alpha/2, n-1}^2 \right\} = 1 - \alpha$$

Therefore, a significance level α test is to

$$\begin{array}{ll} \text{accept } H_0 & \text{if } \chi_{1-\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi_{\alpha/2, n-1}^2 \\ \text{reject } H_0 & \text{otherwise} \end{array}$$

The preceding test can be implemented by first computing the value of the test statistic $(n-1)S^2/\sigma_0^2$ — call it c . Then compute the probability that a chi-square random variable with $n-1$ degrees of freedom would be (a) less than and (b) greater than c . If either of these probabilities is less than $\alpha/2$, then the hypothesis is rejected. In other words, the p -value of the test data is

$$p\text{-value} = 2 \min(P\{\chi_{n-1}^2 < c\}, 1 - P\{\chi_{n-1}^2 < c\})$$

The quantity $P\{\chi_{n-1}^2 < c\}$ can be obtained from Program 5.8.1.A. The p -value for a one-sided test is similarly obtained.

EXAMPLE 8.5a A machine that automatically controls the amount of ribbon on a tape has recently been installed. This machine will be judged to be effective if the standard deviation σ of the amount of ribbon on a tape is less than .15 cm. If a sample of 20 tapes yields a sample variance of $S^2 = .025 \text{ cm}^2$, are we justified in concluding that the machine is ineffective?

SOLUTION We will test the hypothesis that the machine is effective, since a rejection of this hypothesis will then enable us to conclude that it is ineffective. Since we are thus interested in testing

$$H_0 : \sigma^2 \leq .0225 \quad \text{versus} \quad H_1 : \sigma^2 > .0225$$

it follows that we would want to reject H_0 when S^2 is large. Hence, the p -value of the preceding test data is the probability that a chi-square random variable with 19 degrees of freedom would exceed the observed value of $19S^2/.0225 = 19 \times .025/.0225 = 21.111$. That is,

$$\begin{aligned} p\text{-value} &= P\{\chi_{19}^2 > 21.111\} \\ &= 1 - .6693 = .3307 \quad \text{from Program 5.8.1.A} \end{aligned}$$

Therefore, we must conclude that the observed value of $S^2 = .025$ is not large enough to reasonably preclude the possibility that $\sigma^2 \leq .0225$, and so the null hypothesis is accepted. ■

8.5.1 TESTING FOR THE EQUALITY OF VARIANCES OF TWO NORMAL POPULATIONS

Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ denote independent samples from two normal populations having respective (unknown) parameters μ_x, σ_x^2 and μ_y, σ_y^2 and consider a test of

$$H_0 : \sigma_x^2 = \sigma_y^2 \quad \text{versus} \quad H_1 : \sigma_x^2 \neq \sigma_y^2$$

If we let

$$\begin{aligned} S_x^2 &= \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} \\ S_y^2 &= \frac{\sum_{i=1}^m (Y_i - \bar{Y})^2}{m-1} \end{aligned}$$

denote the sample variances, then as shown in Section 6.5, $(n-1)S_x^2/\sigma_x^2$ and $(m-1)S_y^2/\sigma_y^2$ are independent chi-square random variables with $n-1$ and $m-1$ degrees of freedom, respectively. Therefore, $(S_x^2/\sigma_x^2)/(S_y^2/\sigma_y^2)$ has an F -distribution with parameters $n-1$ and $m-1$. Hence, when H_0 is true

$$S_x^2/S_y^2 \sim F_{n-1, m-1}$$

and so

$$P_{H_0}\{F_{1-\alpha/2, n-1, m-1} \leq S_x^2/S_y^2 \leq F_{\alpha/2, n-1, m-1}\} = 1 - \alpha$$

Thus, a significance level α test of H_0 against H_1 is to

$$\begin{array}{ll} \text{accept } H_0 & \text{if } F_{1-\alpha/2, n-1, m-1} < S_x^2/S_y^2 < F_{\alpha/2, n-1, m-1} \\ \text{reject } H_0 & \text{otherwise} \end{array}$$

The preceding test can be effected by first determining the value of the test statistic S_x^2/S_y^2 , say its value is v , and then computing $P\{F_{n-1, m-1} \leq v\}$ where $F_{n-1, m-1}$ is an F -random variable with parameters $n-1$, $m-1$. If this probability is either less than $\alpha/2$ (which occurs when S_x^2 is significantly less than S_y^2) or greater than $1 - \alpha/2$ (which occurs when S_x^2 is significantly greater than S_y^2), then the hypothesis is rejected. In other words, the p -value of the test data is

$$p\text{-value} = 2 \min(P\{F_{n-1, m-1} < v\}, 1 - P\{F_{n-1, m-1} < v\})$$

The test now calls for rejection whenever the significance level α is at least as large as the p -value.

EXAMPLE 8.5b There are two different choices of a catalyst to stimulate a certain chemical process. To test whether the variance of the yield is the same no matter which catalyst is used, a sample of 10 batches is produced using the first catalyst, and 12 using the second. If the resulting data are $S_1^2 = .14$ and $S_2^2 = .28$, can we reject, at the 5 percent level, the hypothesis of equal variance?

SOLUTION Program 5.8.3, which computes the F cumulative distribution function, yields that

$$P\{F_{9,11} \leq .5\} = .1539$$

Hence,

$$\begin{aligned} p\text{-value} &= 2 \min\{.1539, .8461\} \\ &= .3074 \end{aligned}$$

and so the hypothesis of equal variance cannot be rejected. ■

8.6 HYPOTHESIS TESTS IN BERNOULLI POPULATIONS

The binomial distribution is frequently encountered in engineering problems. For a typical example, consider a production process that manufactures items that can be classified in one of two ways — either as acceptable or as defective. An assumption often made is that each item produced will, independently, be defective with probability p , and so the number of defects in a sample of n items will thus have a binomial distribution with parameters (n, p) . We will now consider a test of

$$H_0 : p \leq p_0 \quad \text{versus} \quad H_1 : p > p_0$$

where p_0 is some specified value.

If we let X denote the number of defects in the sample of size n , then it is clear that we wish to reject H_0 when X is large. To see how large it needs to be to justify rejection at the α level of significance, note that

$$P\{X \geq k\} = \sum_{i=k}^n P\{X = i\} = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Now it is certainly intuitive (and can be proven) that $P\{X \geq k\}$ is an increasing function of p — that is, the probability that the sample will contain at least k errors increases in the defect probability p . Using this, we see that when H_0 is true (and so $p \leq p_0$),

$$P\{X \geq k\} \leq \sum_{i=k}^n \binom{n}{i} p_0^i (1-p_0)^{n-i}$$

Hence, a significance level α test of $H_0 : p \leq p_0$ versus $H_1 : p > p_0$ is to reject H_0 when

$$X \geq k^*$$

where k^* is the smallest value of k for which $\sum_{i=k}^n \binom{n}{i} p_0^i (1-p_0)^{n-i} \leq \alpha$. That is,

$$k^* = \min \left\{ k : \sum_{i=k}^n \binom{n}{i} p_0^i (1-p_0)^{n-i} \leq \alpha \right\}$$

This test can best be performed by first determining the value of the test statistic — say, $X = x$ — and then computing the p -value given by

$$\begin{aligned} p\text{-value} &= P\{B(n, p_0) \geq x\} \\ &= \sum_{i=x}^n \binom{n}{i} p_0^i (1-p_0)^{n-i} \end{aligned}$$

EXAMPLE 8.6a A computer chip manufacturer claims that no more than 2 percent of the chips it sends out are defective. An electronics company, impressed with this claim, has purchased a large quantity of such chips. To determine if the manufacturer's claim can be taken literally, the company has decided to test a sample of 300 of these chips. If 10 of these 300 chips are found to be defective, should the manufacturer's claim be rejected?

SOLUTION Let us test the claim at the 5 percent level of significance. To see if rejection is called for, we need to compute the probability that the sample of size 300 would have resulted in 10 or more defectives when p is equal to .02. (That is, we compute the p -value.) If this probability is less than or equal to .05, then the manufacturer's claim should be rejected. Now

$$\begin{aligned}
P_{.02}\{X \geq 10\} &= 1 - P_{.02}\{X < 10\} \\
&= 1 - \sum_{i=0}^9 \binom{300}{i} (.02)^i (.98)^{300-i} \\
&= .0818 \quad \text{from Program 3.1}
\end{aligned}$$

and so the manufacturer's claim cannot be rejected at the 5 percent level of significance. ■

EXAMPLE 8.6b In an attempt to show that proofreader A is superior to proofreader B, both proofreaders were given the same manuscript to read. If proofreader A found 28 errors, and proofreader B found 18, with 10 of these errors being found by both, can we conclude that A is the superior proofreader?

SOLUTION To begin note that A found 18 errors that B missed, and that B found 8 that A missed. Hence, a total of 26 errors were found by just a single proofreader. Now, if A and B were equally competent then they would be equally likely to be the sole finder of an error found by just one of them. Consequently, if A and B were equally competent then each of the 26 singly found errors would have been found by A with probability 1/2. Hence, to establish that A is the superior proofreader the result of 18 successes in 26 trials must be strong enough to reject the null hypothesis when testing

$$H_0 : p \leq 1/2 \quad \text{versus} \quad H_1 : p > 1/2$$

where p is a Bernoulli probability that a trial is a success. Because the resultant p-value for the data cited is

$$p\text{-value} = P\{\text{Bin}(26, .5) \geq 18\} = .0378$$

the null hypothesis would be rejected at the 5 percent level of significance, thus enabling one to conclude (at that level of significance) that A is the superior proofreader. ■

When the sample size n is large, we can derive an *approximate* significance level α test of $H_0 : p \leq p_0$ versus $H_1 : p > p_0$ by using the normal approximation to the binomial. It works as follows: Because when n is large X will have approximately a normal distribution with mean and variance

$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

it follows that

$$\frac{X - np}{\sqrt{np(1 - p)}}$$

will have approximately a standard normal distribution. Therefore, an approximate significance level α test would be to reject H_0 if

$$\frac{X - np_0}{\sqrt{np_0(1 - p_0)}} \geq z_\alpha$$

Equivalently, one can use the normal approximation to approximate the p -value.

EXAMPLE 8.6c In Example 8.6a, $np_0 = 300(.02) = 6$, and $\sqrt{np_0(1 - p_0)} = \sqrt{5.88}$. Consequently, the p -value that results from the data $X = 10$ is

$$\begin{aligned} p\text{-value} &= P_{.02}\{X \geq 10\} \\ &= P_{.02}\{X \geq 9.5\} \\ &= P_{.02}\left\{\frac{X - 6}{\sqrt{5.88}} \geq \frac{9.5 - 6}{\sqrt{5.88}}\right\} \\ &\approx P\{Z \geq 1.443\} \\ &= .0745 \end{aligned}$$

Thus, whereas the exact p -value is .0818, the normal approximation gives the value .0745. ■

Suppose now that we want to test the null hypothesis that p is equal to some specified value; that is, we want to test

$$H_0 : p = p_0 \quad \text{versus} \quad H_1 : p \neq p_0$$

If X , a binomial random variable with parameters n and p , is observed to equal x , then a significance level α test would reject H_0 if the value x was either significantly larger or significantly smaller than what would be expected when p is equal to p_0 . More precisely, the test would reject H_0 if either

$$P\{\text{Bin}(n, p_0) \geq x\} \leq \alpha/2 \quad \text{or} \quad P\{\text{Bin}(n, p_0) \leq x\} \leq \alpha/2$$

In other words, the p -value when $X = x$ is

$$p\text{-value} = 2 \min(P\{\text{Bin}(n, p_0) \geq x\}, P\{\text{Bin}(n, p_0) \leq x\})$$

EXAMPLE 8.6d Historical data indicate that 4 percent of the components produced at a certain manufacturing facility are defective. A particularly acrimonious labor dispute has recently been concluded, and management is curious about whether it will result in any change in this figure of 4 percent. If a random sample of 500 items indicated 16

defectives (3.2 percent), is this significant evidence, at the 5 percent level of significance, to conclude that a change has occurred?

SOLUTION To be able to conclude that a change has occurred, the data need to be strong enough to reject the null hypothesis when we are testing

$$H_0 : p = .04 \quad \text{versus} \quad H_1 : p \neq .04$$

where p is the probability that an item is defective. The p -value of the observed data of 16 defectives in 500 items is

$$p\text{-value} = 2 \min\{P\{X \leq 16\}, P\{X \geq 16\}\}$$

where X is a binomial (500, .04) random variable. Since $500 \times .04 = 20$, we see that

$$p\text{-value} = 2P\{X \leq 16\}$$

Since X has mean 20 and standard deviation $\sqrt{20(.96)} = 4.38$, it is clear that twice the probability that X will be less than or equal to 16 — a value less than one standard deviation lower than the mean — is not going to be small enough to justify rejection. Indeed, it can be shown that

$$p\text{-value} = 2P\{X \leq 16\} = .432$$

and so there is not sufficient evidence to reject the hypothesis that the probability of a defective item has remained unchanged. ■

8.6.1 TESTING THE EQUALITY OF PARAMETERS IN TWO BERNOULLI POPULATIONS

Suppose there are two distinct methods for producing a certain type of chip; and suppose that chips produced by the first method will, independently, be defective with probability p_1 , with the corresponding probability being p_2 for those produced by the second method. To test the hypothesis that $p_1 = p_2$, a sample of n_1 chips is produced using method 1 and n_2 using method 2.

Let X_1 denote the number of defective chips obtained from the first sample and X_2 for the second. Thus, X_1 and X_2 are independent binomial random variables with respective parameters (n_1, p_1) and (n_2, p_2) . Suppose that $X_1 + X_2 = k$ and so there have been a total of k defectives. Now, if H_0 is true, then each of the $n_1 + n_2$ chips produced will have the same probability of being defective, and so the determination of the k defectives will have the same distribution as a random selection of a sample of size k from a population of $n_1 + n_2$ items of which n_1 are white and n_2 are black. In other words, given a total of k defectives, the conditional distribution of the number of defective chips

obtained from method 1 will, when H_0 is true, have the following hypergeometric distribution* :

$$P_{H_0}\{X_1 = i | X_1 + X_2 = k\} = \frac{\binom{n_1}{i} \binom{n_2}{k-i}}{\binom{n_1+n_2}{k}}, \quad i = 0, 1, \dots, k \quad (8.6.1)$$

Now, in testing

$$H_0 : p_1 = p_2 \quad \text{versus} \quad H_1 : p_1 \neq p_2$$

it seems reasonable to reject the null hypothesis when the proportion of defective chips produced by method 1 is much different from the proportion of defectives obtained under method 2. Therefore, if there is a total of k defectives, then we would expect, when H_0 is true, that X_1/n_1 (the proportion of defective chips produced by method 1) would be close to $(k - X_1)/n_2$ (the proportion of defective chips produced by method 2). Because X_1/n_1 and $(k - X_1)/n_2$ will be farthest apart when X_1 is either very small or very large, it thus seems that a reasonable significance level α test of Equation 8.6.1 is as follows. If $X_1 + X_2 = k$, then one should

$$\begin{array}{ll} \text{reject } H_0 & \text{if either } P\{X \leq x_1\} \leq \alpha/2 \quad \text{or} \quad P\{X \geq x_1\} \leq \alpha/2 \\ \text{accept } H_0 & \text{otherwise} \end{array}$$

where X is a hypergeometric random variable with probability mass function

$$P\{X = i\} = \frac{\binom{n_1}{i} \binom{n_2}{k-i}}{\binom{n_1+n_2}{k}} \quad i = 0, 1, \dots, k \quad (8.6.2)$$

In other words, this test will call for rejection if the significance level is at least as large as the p -value given by

$$p\text{-value} = 2 \min(P\{X \leq x_1\}, P\{X \geq x_1\}) \quad (8.6.3)$$

This is called the *Fisher-Irwin test*.

* See Example 5.3b for a formal verification of Equation 8.6.1.

COMPUTATIONS FOR THE FISHER-IRWIN TEST

To utilize the Fisher-Irwin test, we need to be able to compute the hypergeometric distribution function. To do so, note that with X having mass function Equation 8.6.2,

$$\frac{P\{X = i + 1\}}{P\{X = i\}} = \frac{\binom{n_1}{i+1} \binom{n_2}{k-i-1}}{\binom{n_1}{i} \binom{n_2}{k-i}} \quad (8.6.4)$$

$$= \frac{(n_1 - i)(k - i)}{(i + 1)(n_2 - k + i + 1)} \quad (8.6.5)$$

where the verification of the final equality is left as an exercise.

Program 8.6.1 uses the preceding identity to compute the p -value of the data for the Fisher-Irwin test of the equality of two Bernoulli probabilities. The program will work best if the Bernoulli outcome that is called unsuccessful (or defective) is the one whose probability is less than .5. For instance, if over half the items produced are defective, then rather than testing that the defect probability is the same in both samples, one should test that the probability of producing an acceptable item is the same in both samples.

EXAMPLE 8.6e Suppose that method 1 resulted in 20 unacceptable transistors out of 100 produced, whereas method 2 resulted in 12 unacceptable transistors out of 100 produced. Can we conclude from this, at the 10 percent level of significance, that the two methods are equivalent?

SOLUTION Upon running Program 8.6.1, we obtain that

$$p\text{-value} = .1763$$

Hence, the hypothesis that the two methods are equivalent cannot be rejected. ■

The ideal way to test the hypothesis that the results of two different treatments are identical is to randomly divide a group of people into a set that will receive the first treatment and one that will receive the second. However, such randomization is not always possible. For instance, if we want to study whether drinking alcohol increases the risk of prostate cancer, we cannot instruct a randomly chosen sample to drink alcohol. An alternative way to study the hypothesis is to use an *observational* study that begins by randomly choosing a set of drinkers and one of nondrinkers. These sets are followed for a period of time and the resulting data are then used to test the hypothesis that members of the two groups have the same risk for prostate cancer.

Our next sample illustrates another way of performing an observational study.

EXAMPLE 8.6f In 1970, the researchers Herbst, Ulfelder, and Poskanzer (H-U-P) suspected that vaginal cancer in young women, a rather rare disease, might be caused by

one's mother having taken the drug diethylstilbestrol (usually referred to as DES) while pregnant. To study this possibility, the researchers could have performed an observational study by searching for a (treatment) group of women whose mothers took DES when pregnant and a (control) group of women whose mothers did not. They could then observe these groups for a period of time and use the resulting data to test the hypothesis that the probabilities of contracting vaginal cancer are the same for both groups. However, because vaginal cancer is so rare (in both groups) such a study would require a large number of individuals in both groups and would probably have to continue for many years to obtain significant results. Consequently, H-U-P decided on a different type of observational study. They uncovered 8 women between the ages of 15 and 22 who had vaginal cancer. Each of these women (called cases) was then matched with 4 others, called *referents or controls*. Each of the referents of a case was free of the cancer and was born within 5 days in the same hospital and in the same type of room (either private or public) as the case. Arguing that if DES had no effect on vaginal cancer then the probability, call it p_c , that the mother of a case took DES would be the same as the probability, call it p_r , that the mother of a referent took DES, the researchers H-U-P decided to test

$$H_0 : p_c = p_r \quad \text{against} \quad H_1 : p_c \neq p_r$$

Discovering that 7 of the 8 cases had mothers who took DES while pregnant, while none of the 32 referents had mothers who took the drug, the researchers (see Herbst, A., Ulfelder, H., and Poskanzer, D., "Adenocarcinoma of the Vagina: Association of Maternal Stilbestrol Therapy with Tumor Appearance in Young Women," *New England Journal of Medicine*, **284**, 878–881, 1971) concluded that there was a strong association between DES and vaginal cancer. (The p -value for these data is approximately 0.) ■

When n_1 and n_2 are large, an approximate level α test of $H_0 : p_1 = p_2$, based on the normal approximation to the binomial, is outlined in Problem 63.

8.7 TESTS CONCERNING THE MEAN OF A POISSON DISTRIBUTION

Let X denote a Poisson random variable having mean λ and consider a test of

$$H_0 : \lambda = \lambda_0 \quad \text{versus} \quad H_1 : \lambda \neq \lambda_0$$

If the observed value of X is $X = x$, then a level α test would reject H_0 if either

$$P_{\lambda_0}\{X \geq x\} \leq \alpha/2 \quad \text{or} \quad P_{\lambda_0}\{X \leq x\} \leq \alpha/2 \quad (8.7.1)$$

where P_{λ_0} means that the probability is computed under the assumption that the Poisson mean is λ_0 . It follows from Equation 8.7.1 that the p -value is given by

$$p\text{-value} = 2 \min(P_{\lambda_0}\{X \geq x\}, P_{\lambda_0}\{X \leq x\})$$

The calculation of the preceding probabilities that a Poisson random variable with mean λ_0 is greater (less) than or equal to x can be obtained by using Program 5.2.

EXAMPLE 8.7a Management's claim that the mean number of defective computer chips produced daily is not greater than 25 is in dispute. Test this hypothesis, at the 5 percent level of significance, if a sample of 5 days revealed 28, 34, 32, 38, and 22 defective chips.

SOLUTION Because each individual computer chip has a very small chance of being defective, it is probably reasonable to suppose that the daily number of defective chips is approximately a Poisson random variable, with mean, say, λ . To see whether or not the manufacturer's claim is credible, we shall test the hypothesis

$$H_0 : \lambda \leq 25 \quad \text{versus} \quad H_1 : \lambda > 25$$

Now, under H_0 , the total number of defective chips produced over a 5-day period is Poisson distributed (since the sum of independent Poisson random variables is Poisson) with a mean no greater than 125. Since this number is equal to 154, it follows that the p -value of the data is given by

$$\begin{aligned} p\text{-value} &= P_{125}\{X \geq 154\} \\ &= 1 - P_{125}\{X \leq 153\} \\ &= .0066 \quad \text{from Program 5.2} \end{aligned}$$

Therefore, the manufacturer's claim is rejected at the 5 percent (as it would be even at the 1 percent) level of significance. ■

REMARK

If Program 5.2 is not available, one can use the fact that a Poisson random variable with mean λ is, for large λ , approximately normally distributed with a mean and variance equal to λ .

8.7.1 TESTING THE RELATIONSHIP BETWEEN TWO POISSON PARAMETERS

Let X_1 and X_2 be independent Poisson random variables with respective means λ_1 and λ_2 , and consider a test of

$$H_0 : \lambda_2 = c\lambda_1 \quad \text{versus} \quad H_1 : \lambda_2 \neq c\lambda_1$$

for a given constant c . Our test of this is a conditional test (similar in spirit to the Fisher-Irwin test of Section 8.6.1), which is based on the fact that the conditional distribution of X_1 given the sum of X_1 and X_2 is binomial. More specifically, we have the following proposition.

PROPOSITION 8.7.1

$$P\{X_1 = k | X_1 + X_2 = n\} = \binom{n}{k} [\lambda_1 / (\lambda_1 + \lambda_2)]^k [\lambda_2 / (\lambda_1 + \lambda_2)]^{n-k}$$

Proof

$$\begin{aligned} P\{X_1 = k | X_1 + X_2 = n\} &= \frac{P\{X_1 = k, X_1 + X_2 = n\}}{P\{X_1 + X_2 = n\}} \\ &= \frac{P\{X_1 = k, X_2 = n - k\}}{P\{X_1 + X_2 = n\}} \\ &= \frac{P\{X_1 = k\}P\{X_2 = n - k\}}{P\{X_1 + X_2 = n\}} \quad \text{by independence} \\ &= \frac{\exp\{-\lambda_1\} \lambda_1^k / k! \exp\{-\lambda_2\} \lambda_2^{n-k} / (n-k)!}{\exp\{-(\lambda_1 + \lambda_2)\} (\lambda_1 + \lambda_2)^n / n!} \\ &= \frac{n!}{(n-k)!k!} [\lambda_1 / (\lambda_1 + \lambda_2)]^k [\lambda_2 / (\lambda_1 + \lambda_2)]^{n-k} \end{aligned}$$

where the next to last equality follows because the sum of independent Poisson random variables is also Poisson. ■

It follows from Proposition 8.7.1 that, if H_0 is true, then the conditional distribution of X_1 given that $X_1 + X_2 = n$ is the binomial distribution with parameters n and $p = 1/(1 + c)$. From this we can conclude that if $X_1 + X_2 = n$, then H_0 should be rejected if the observed value of X_1 , call it x_1 , is such that either

$$P\{\text{Bin}(n, 1/(1 + c)) \geq x_1\} \leq \alpha/2$$

or

$$P\{\text{Bin}(n, 1/(1 + c)) \leq x_1\} \leq \alpha/2$$

EXAMPLE 8.7b An industrial concern runs two large plants. If the number of accidents during the past 8 weeks at plant 1 were 16, 18, 9, 22, 17, 19, 24, 8 while the number of accidents during the last 6 weeks at plant 2 were 22, 18, 26, 30, 25, 28, can we conclude, at the 5 percent level of significance, that the safety conditions differ from plant to plant?

SOLUTION Since there is a small probability of an industrial accident in any given minute, it would seem that the weekly number of such accidents should have approximately a Poisson distribution. If we let X_1 denote the total number of accidents during an 8-week period at plant 1, and let X_2 be the number during a 6-week period at plant 2, then if the safety conditions did not differ at the two plants we would have that

$$\lambda_2 = \frac{3}{4}\lambda_1$$

where $\lambda_i \equiv E[X_i]$, $i = 1, 2$. Hence, as $X_1 = 133$, $X_2 = 149$ it follows that the p -value of the test of

$$H_0 : \lambda_2 = \frac{3}{4}\lambda_1 \quad \text{versus} \quad H_1 : \lambda_2 \neq \frac{3}{4}\lambda_1$$

is given by

$$\begin{aligned} p\text{-value} &= 2 \min\left(P\{\text{Bin}(282, \frac{4}{7}) \geq 133\}, P\{\text{Bin}(282, \frac{4}{7}) \leq 133\}\right) \\ &= 9.408 \times 10^{-4} \end{aligned}$$

Thus, the hypothesis that the safety conditions at the two plants are equivalent is rejected. ■

Problems

1. Consider a trial in which a jury must decide between the hypothesis that the defendant is guilty and the hypothesis that he or she is innocent.
 - (a) In the framework of hypothesis testing and the U.S. legal system, which of the hypotheses should be the null hypothesis?
 - (b) What do you think would be an appropriate significance level in this situation?
2. A colony of laboratory mice consists of several thousand mice. The average weight of all the mice is 32 grams with a standard deviation of 4 grams. A laboratory assistant was asked by a scientist to select 25 mice for an experiment. However, before performing the experiment the scientist decided to weigh the mice as an indicator of whether the assistant's selection constituted a random sample or whether it was made with some unconscious bias (perhaps the mice selected were the ones that were slowest in avoiding the assistant, which might indicate some inferiority about this group). If the sample mean of the

25 mice was 30.4, would this be significant evidence, at the 5 percent level of significance, against the hypothesis that the selection constituted a random sample?

3. A population distribution is known to have standard deviation 20. Determine the p -value of a test of the hypothesis that the population mean is equal to 50, if the average of a sample of 64 observations is
(a) 52.5; (b) 55.0; (c) 57.5.
4. In a certain chemical process, it is very important that a particular solution that is to be used as a reactant have a pH of exactly 8.20. A method for determining pH that is available for solutions of this type is known to give measurements that are normally distributed with a mean equal to the actual pH and with a standard deviation of .02. Suppose 10 independent measurements yielded the following pH values:

8.18	8.17
8.16	8.15
8.17	8.21
8.22	8.16
8.19	8.18

- (a) What conclusion can be drawn at the $\alpha = .10$ level of significance?
- (b) What about at the $\alpha = .05$ level of significance?
5. The mean breaking strength of a certain type of fiber is required to be at least 200 psi. Past experience indicates that the standard deviation of breaking strength is 5 psi. If a sample of 8 pieces of fiber yielded breakage at the following pressures,

210	198
195	202
197.4	196
199	195.5

would you conclude, at the 5 percent level of significance, that the fiber is unacceptable? What about at the 10 percent level of significance?

6. It is known that the average height of a man residing in the United States is 5 feet 10 inches and the standard deviation is 3 inches. To test the hypothesis that men in your city are “average,” a sample of 20 men have been chosen. The heights of the men in the sample follow:

Man	Height in	Inches	Man
1	72	70.4	11
2	68.1	76	12
3	69.2	72.5	13
4	72.8	74	14
5	71.2	71.8	15
6	72.2	69.6	16
7	70.8	75.6	17
8	74	70.6	18
9	66	76.2	19
10	70.3	77	20

What do you conclude? Explain what assumptions you are making.

7. Suppose in Problem 4 that we wished to design a test so that if the pH were really equal to 8.20, then this conclusion will be reached with probability equal to .95. On the other hand, if the pH differs from 8.20 by .03 (in either direction), we want the probability of picking up such a difference to exceed .95.
 - (a) What test procedure should be used?
 - (b) What is the required sample size?
 - (c) If $\bar{x} = 8.31$, what is your conclusion?
 - (d) If the actual pH is 8.32, what is the probability of concluding that the pH is not 8.20, using the foregoing procedure?
8. Verify that the approximation in Equation 8.3.7 remains valid even when $\mu_1 < \mu_0$.
9. A British pharmaceutical company, Glaxo Holdings, has recently developed a new drug for migraine headaches. Among the claims Glaxo made for its drug, called sumatriptan, was that the mean time it takes for it to enter the bloodstream is less than 10 minutes. To convince the Food and Drug Administration of the validity of this claim, Glaxo conducted an experiment on a randomly chosen set of migraine sufferers. To prove its claim, what should they have taken as the null and what as the alternative hypothesis?
10. The weights of salmon grown at a commercial hatchery are normally distributed with a standard deviation of 1.2 pounds. The hatchery claims that the mean weight of this year's crop is at least 7.6 pounds. Suppose a random sample of 16 fish yielded an average weight of 7.2 pounds. Is this strong enough evidence to reject the hatchery's claims at the
 - (a) 5 percent level of significance;
 - (b) 1 percent level of significance?
 - (c) What is the p -value?

11. Consider a test of $H_0 : \mu \leq 100$ versus $H_1 : \mu > 100$. Suppose that a sample of size 20 has a sample mean of $\bar{X} = 105$. Determine the p -value of this outcome if the population standard deviation is known to equal
(a) 5; (b) 10; (c) 15.
12. An advertisement for a new toothpaste claims that it reduces cavities of children in their cavity-prone years. Cavities per year for this age group are normal with mean 3 and standard deviation 1. A study of 2,500 children who used this toothpaste found an average of 2.95 cavities per child. Assume that the standard deviation of the number of cavities of a child using this new toothpaste remains equal to 1.
 - (a) Are these data strong enough, at the 5 percent level of significance, to establish the claim of the toothpaste advertisement?
 - (b) Do the data convince you to switch to this new toothpaste?
13. There is some variability in the amount of phenobarbital in each capsule sold by a manufacturer. However, the manufacturer claims that the mean value is 20.0 mg. To test this, a sample of 25 pills yielded a sample mean of 19.7 with a sample standard deviation of 1.3. What inference would you draw from these data? In particular, are the data strong enough evidence to discredit the claim of the manufacturer? Use the 5 percent level of significance.
14. Twenty years ago, entering male high school students of Central High could do an average of 24 pushups in 60 seconds. To see whether this remains true today, a random sample of 36 freshmen was chosen. If their average was 22.5 with a sample standard deviation of 3.1, can we conclude that the mean is no longer equal to 24? Use the 5 percent level of significance.
15. The mean response time of a species of pigs to a stimulus is .8 seconds. Twenty-eight pigs were given 2 oz of alcohol and then tested. If their average response time was 1.0 seconds with a standard deviation of .3 seconds, can we conclude that alcohol affects the mean response time? Use the 5 percent level of significance.
16. Suppose that team A and team B are to play a National Football League game and that team A is favored by f points. Let $S(A)$ and $S(B)$ denote the scores of teams A and B , and let $X = S(A) - S(B) - f$. That is, X is the amount by which team A beats the point spread. It has been claimed that the distribution of X is normal with mean 0 and standard deviation 14. Use data from randomly chosen football games to test this hypothesis.
17. A medical scientist believes that the average basal temperature of (outwardly) healthy individuals has increased over time and is now greater than 98.6 degrees Fahrenheit (37 degrees Celsius). To prove this, she has randomly selected 100 healthy individuals. If their mean temperature is 98.74 with a sample standard deviation of 1.1 degrees, does this prove her claim at the 5 percent level? What about at the 1 percent level?

18. Use the results of a Sunday's worth of NFL professional football games to test the hypothesis that the average number of points scored by winning teams is less than or equal to 28. Use the 5 percent level of significance.
19. Use the results of a Sunday's worth of major league baseball scores to test the hypothesis that the average number of runs scored by winning teams is at least 5.6. Use the 5 percent level of significance.
20. A car is advertised as having a gas mileage rating of at least 30 miles/gallon in highway driving. If the miles per gallon obtained in 10 independent experiments are 26, 24, 20, 25, 27, 25, 28, 30, 26, 33, should you believe the advertisement? What assumptions are you making?
21. A producer specifies that the mean lifetime of a certain type of battery is at least 240 hours. A sample of 18 such batteries yielded the following data.

237	242	232
242	248	230
244	243	254
262	234	220
225	236	232
218	228	240

Assuming that the life of the batteries is approximately normally distributed, do the data indicate that the specifications are not being met?

22. Use the data of Example 2.3i of Chapter 2 to test the null hypothesis that the average noise level directly outside of Grand Central Station is less than or equal to 80 decibels.
23. An oil company claims that the sulfur content of its diesel fuel is at most .15 percent. To check this claim, the sulfur contents of 40 randomly chosen samples were determined; the resulting sample mean and sample standard deviation were .162 and .040. Using the 5 percent level of significance, can we conclude that the company's claims are invalid?
24. A company supplies plastic sheets for industrial use. A new type of plastic has been produced and the company would like to claim that the average stress resistance of this new product is at least 30.0, where stress resistance is measured in pounds per square inch (psi) necessary to crack the sheet. The following random sample was drawn off the production line. Based on this sample, would the claim clearly be unjustified?

30.1	32.7	22.5	27.5
27.7	29.8	28.9	31.4
31.2	24.3	26.4	22.8
29.1	33.4	32.5	21.7

Assume normality and use the 5 percent level of significance.

25. It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of current gain is 200 with a sample standard deviation of 35, would the claim be rejected at the 5 percent level of significance if
- (a) the sample size is 25;
 - (b) the sample size is 64?

26. A manufacturer of capacitors claims that the breakdown voltage of these capacitors has a mean value of at least 100 V. A test of 12 of these capacitors yielded the following breakdown voltages:

96, 98, 105, 92, 111, 114, 99, 103, 95, 101, 106, 97

Do these results prove the manufacturer's claim? Do they disprove them?

27. A sample of 10 fish were caught at lake A and their PCB concentrations were measured using a certain technique. The resulting data in parts per million were

Lake A: 11.5, 10.8, 11.6, 9.4, 12.4, 11.4, 12.2, 11, 10.6, 10.8

In addition, a sample of 8 fish were caught at lake B and their levels of PCB were measured by a different technique than that used at lake A. The resultant data were

Lake B: 11.8, 12.6, 12.2, 12.5, 11.7, 12.1, 10.4, 12.6

If it is known that the measuring technique used at lake A has a variance of .09 whereas the one used at lake B has a variance of .16, could you reject (at the 5 percent level of significance) a claim that the two lakes are equally contaminated?

28. A method for measuring the pH level of a solution yields a measurement value that is normally distributed with a mean equal to the actual pH of the solution and with a standard deviation equal to .05. An environmental pollution scientist claims that two different solutions come from the same source. If this were so, then the pH level of the solutions would be equal. To test the plausibility of this claim, 10 independent measurements were made of the pH level for both solutions, with the following data resulting.

Measurements of Solution A	Measurements of Solution B
6.24	6.27
6.31	6.25
6.28	6.33
6.30	6.27
6.25	6.24
6.26	6.31
6.24	6.28
6.29	6.29
6.22	6.34
6.28	6.27

- (a) Do the data disprove the scientist's claim? Use the 5 percent level of significance.
 (b) What is the p -value?

29. The following are the values of independent samples from two different populations.

Sample 1	122, 114, 130, 165, 144, 133, 139, 142, 150
Sample 2	108, 125, 122, 140, 132, 120, 137, 128, 138

Let μ_1 and μ_2 be the respective means of the two populations. Find the p -value of the test of the null hypothesis

$$H_0 : \mu_1 \leq \mu_2$$

versus the alternative

$$H_1 : \mu_1 > \mu_2$$

when the population standard deviations are $\sigma_1 = 10$ and
 (a) $\sigma_2 = 5$; (b) $\sigma_2 = 10$; (c) $\sigma_2 = 20$.

30. The data below give the lifetimes in hundreds of hours of samples of two types of electronic tubes. Past lifetime data of such tubes have shown that they can often be modeled as arising from a lognormal distribution. That is, the logarithms of the data are normally distributed. Assuming that variance of the logarithms is equal

for the two populations, test, at the 5 percent level of significance, the hypothesis that the two population distributions are identical.

Type 1	32, 84, 37, 42, 78, 62, 59, 74
Type 2	39, 111, 55, 106, 90, 87, 85

31. The viscosity of two different brands of car oil is measured and the following data resulted:

Brand 1	10.62, 10.58, 10.33, 10.72, 10.44, 10.74
Brand 2	10.50, 10.52, 10.58, 10.62, 10.55, 10.51, 10.53

Test the hypothesis that the mean viscosity of the two brands is equal, assuming that the populations have normal distributions with equal variances.

32. It is argued that the resistance of wire A is greater than the resistance of wire B. You make tests on each wire with the following results.

Wire A	Wire B
.140 ohm	.135 ohm
.138	.140
.143	.136
.142	.142
.144	.138
.137	.140

What conclusion can you draw at the 10 percent significance level? Explain what assumptions you are making.

In Problems 33 through 40, assume that the population distributions are normal and have equal variances.

33. Twenty-five men between the ages of 25 and 30, who were participating in a well-known heart study carried out in Framingham, Massachusetts, were randomly selected. Of these, 11 were smokers and 14 were not. The following data refer to readings of their systolic blood pressure.

Smokers	Nonsmokers
124	130
134	122
136	128
125	129
133	118
127	122
135	116
131	127
133	135
125	120
118	122
	120
	115
	123

Use these data to test the hypothesis that the mean blood pressures of smokers and nonsmokers are the same.

34. In a 1943 experiment (Whitlock and Bliss, "A Bioassay Technique for Anti-helminthics," *Journal of Parasitology*, **29**, pp. 48–58) 10 albino rats were used to study the effectiveness of carbon tetrachloride as a treatment for worms. Each rat received an injection of worm larvae. After 8 days, the rats were randomly divided into two groups of 5 each; each rat in the first group received a dose of .032 cc of carbon tetrachloride, whereas the dosage for each rat in the second group was .063 cc. Two days later the rats were killed, and the number of adult worms in each rat was determined. The numbers detected in the group receiving the .032 dosage were

421, 462, 400, 378, 413

whereas they were

207, 17, 412, 74, 116

for those receiving the .063 dosage. Do the data prove that the larger dosage is more effective than the smaller?

35. A professor claims that the average starting salary of industrial engineering graduating seniors is greater than that of civil engineering graduates. To study this claim, samples of 16 industrial engineers and 16 civil engineers, all of whom graduated in 2006, were chosen and sample members were queried about their starting salaries. If the industrial engineers had a sample mean salary of \$72,700 and a sample standard deviation of \$2,400, and the civil engineers had a sample mean

salary of \$71,400 and a sample standard deviation of \$2,200, has the professor’s claim been verified? Find the appropriate p -value.

36. In a certain experimental laboratory, a method A for producing gasoline from crude oil is being investigated. Before completing experimentation, a new method B is proposed. All other things being equal, it was decided to abandon A in favor of B only if the average yield of the latter was clearly greater. The yield of both processes is assumed to be normally distributed. However, there has been insufficient time to ascertain their true standard deviations, although there appears to be no reason why they cannot be assumed equal. Cost considerations impose size limits on the size of samples that can be obtained. If a 1 percent significance level is all that is allowed, what would be your recommendation based on the following random samples? The numbers represent percent yield of crude oil.

A	23.2, 26.6, 24.4, 23.5, 22.6, 25.7, 25.5
B	25.7, 27.7, 26.2, 27.9, 25.0, 21.4, 26.1

37. A study was instituted to learn how the diets of women changed during the winter and the summer. A random group of 12 women were observed during the month of July and the percentage of each woman’s calories that came from fat was determined. Similar observations were made on a different randomly selected group of size 12 during the month of January. The results were as follows:

July	32.2, 27.4, 28.6, 32.4, 40.5, 26.2, 29.4, 25.8, 36.6, 30.3, 28.5, 32.0
January	30.5, 28.4, 40.2, 37.6, 36.5, 38.8, 34.7, 29.5, 29.7, 37.2, 41.5, 37.0

Test the hypothesis that the mean fat percentage intake is the same for both months. Use the (a) 5 percent level of significance and (b) 1 percent level of significance.

38. To learn about the feeding habits of bats, 22 bats were tagged and tracked by radio. Of these 22 bats, 12 were female and 10 were male. The distances flown (in meters) between feedings were noted for each of the 22 bats, and the following summary statistics were obtained.

Female Bats	Male Bats
$n = 12$	$m = 10$
$\bar{X} = 180$	$\bar{Y} = 136$
$S_x = 92$	$S_y = 86$

Test the hypothesis that the mean distance flown between feedings is the same for the populations of both male and of female bats. Use the 5 percent level of significance.

39. The following data summary was obtained from a comparison of the lead content of human hair removed from adult individuals that had died between 1880 and 1920 with the lead content of present-day adults. The data are in units of micrograms, equal to one-millionth of a gram.

	1880–1920	Today
Sample size:	30	100
Sample mean:	48.5	26.6
Sample standard deviation:	14.5	12.3

- (a) Do the above data establish, at the 1 percent level of significance, that the mean lead content of human hair is less today than it was in the years between 1880 and 1920? Clearly state what the null and alternative hypotheses are.
- (b) What is the p -value for the hypothesis test in part (a)?
40. Sample weights (in pounds) of newborn babies born in two adjacent counties in western Pennsylvania yielded the following data.

$$\begin{aligned} n &= 53, & m &= 44 \\ \bar{X} &= 6.8, & \bar{Y} &= 7.2 \\ S^2 &= 5.2, & S^2 &= 4.9 \end{aligned}$$

Consider a test of the hypothesis that the mean weight of newborns is the same in both counties. What is the resulting p -value?

41. To verify the hypothesis that blood lead levels tend to be higher for children whose parents work in a factory that uses lead in the manufacturing process, researchers examined lead levels in the blood of 33 children whose parents worked in a battery manufacturing factory (Morton, D., Saah, A., Silberg, S., Owens, W., Roberts, M., and Saah, M., “Lead Absorption in Children of Employees in a Lead-Related Industry,” *American Journal of Epidemiology*, **115**, 549–555, 1982). Each of these children was then *matched* by another child who was of similar age, lived in a similar neighborhood, had a similar exposure to traffic, but whose parent did not work with lead. The blood levels of the 33 cases (sample 1) as well as those of the 33 controls (sample 2) were then used to test the hypothesis that the average blood levels of these groups are the same. If the resulting sample means and sample standard deviations were

$$\bar{x}_1 = .015, \quad s_1 = .004, \quad \bar{x}_2 = .006, \quad s_2 = .006$$

find the resulting p -value. Assume a common variance.

42. Ten pregnant women were given an injection of pitocin to induce labor. Their systolic blood pressures immediately before and after the injection were:

Patient	Before	After	Patient	Before	After
1	134	140	6	140	138
2	122	130	7	118	124
3	132	135	8	127	126
4	130	126	9	125	132
5	128	134	10	142	144

Do the data indicate that injection of this drug changes blood pressure?

43. A question of medical importance is whether jogging leads to a reduction in one's pulse rate. To test this hypothesis, 8 nonjogging volunteers agreed to begin a 1-month jogging program. After the month their pulse rates were determined and compared with their earlier values. If the data are as follows, can we conclude that jogging has had an effect on the pulse rates?

Subject	1	2	3	4	5	6	7	8
Pulse Rate Before	74	86	98	102	78	84	79	70
Pulse Rate After	70	85	90	110	71	80	69	74

44. If $X_1, \dots, X_n$ is a sample from a normal population having unknown parameters μ and σ^2 , devise a significance level α test of

$$H_0 = \sigma^2 \leq \sigma_0^2$$

versus the alternative

$$H_1 = \sigma^2 > \sigma_0^2$$

for a given positive value σ_0^2 .

45. In Problem 44, explain how the test would be modified if the population mean μ were known in advance.
46. A gun-like apparatus has recently been designed to replace needles in administering vaccines. The apparatus can be set to inject different amounts of the serum, but because of random fluctuations the actual amount injected is normally distributed with a mean equal to the setting and with an unknown variance σ^2 . It has been decided that the apparatus would be too dangerous to use if σ exceeds .10. If a random sample of 50 injections resulted in a sample standard deviation of .08, should use of the new apparatus be discontinued? Suppose the level of significance is $\alpha = .10$. Comment on the appropriate choice of a significance level for this problem, as well as the appropriate choice of the null hypothesis.

47. A pharmaceutical house produces a certain drug item whose weight has a standard deviation of .5 milligrams. The company's research team has proposed a new method of producing the drug. However, this entails some costs and will be adopted only if there is strong evidence that the standard deviation of the weight of the items will drop to below .4 milligrams. If a sample of 10 items is produced and has the following weights, should the new method be adopted?

5.728	5.731
5.722	5.719
5.727	5.724
5.718	5.726
5.723	5.722

48. The production of large electrical transformers and capacitors requires the use of polychlorinated biphenyls (PCBs), which are extremely hazardous when released into the environment. Two methods have been suggested to monitor the levels of PCB in fish near a large plant. It is believed that each method will result in a normal random variable that depends on the method. Test the hypothesis at the $\alpha = .10$ level of significance that both methods have the same variance, if a given fish is checked 8 times by each method with the following data (in parts per million) recorded.

Method 1	6.2, 5.8, 5.7, 6.3, 5.9, 6.1, 6.2, 5.7
Method 2	6.3, 5.7, 5.9, 6.4, 5.8, 6.2, 6.3, 5.5

49. In Problem 31, test the hypothesis that the populations have the same variances.
50. If $X_1, \dots, X_n$ is a sample from a normal population with variance σ_x^2 , and $Y_1, \dots, Y_n$ is an independent sample from normal population with variance σ_y^2 , develop a significance level α test of

$$H_0 : \sigma_x^2 < \sigma_y^2 \quad \text{versus} \quad H_1 : \sigma_x^2 > \sigma_y^2$$

51. The amount of surface wax on each side of waxed paper bags is believed to be normally distributed. However, there is reason to believe that there is greater variation in the amount on the inner side of the paper than on the outside. A sample of 75 observations of the amount of wax on each side of these bags is obtained and the following data recorded.

Wax in Pounds per Unit Area of Sample	
Outside Surface	Inside Surface
$\bar{x} = .948$	$\bar{y} = .652$
$\sum x_i^2 = 91$	$\sum y_i^2 = 82$

Conduct a test to determine whether or not the variability of the amount of wax on the inner surface is greater than the variability of the amount on the outer surface ($\alpha = .05$).

52. In a famous experiment to determine the efficacy of aspirin in preventing heart attacks, 22,000 healthy middle-aged men were randomly divided into two equal groups, one of which was given a daily dose of aspirin and the other a placebo that looked and tasted identical to the aspirin. The experiment was halted at a time when 104 men in the aspirin group and 189 in the control group had had heart attacks. Use these data to test the hypothesis that the taking of aspirin does not change the probability of having a heart attack.
53. In the study of Problem 52, it also resulted that 119 from the aspirin group and 98 from the control group suffered strokes. Are these numbers significant to show that taking aspirin changes the probability of having a stroke?
54. A standard drug is known to be effective in 72 percent of the cases in which it is used to treat a certain infection. A new drug has been developed and testing has found it to be effective in 42 cases out of 50. Is this strong enough evidence to prove that the new drug is more effective than the old one? Find the relevant p -value.
55. Three independent news services are running a poll to determine if over half the population supports an initiative concerning limitations on driving automobiles in the downtown area. Each wants to see if the evidence indicates that over half the population is in favor. As a result, all three services will be testing

$$H_0 : p \leq .5 \quad \text{versus} \quad H_1 : p > .5$$

where p is the proportion of the population in favor of the initiative.

- (a) Suppose the first news organization samples 100 people, of which 56 are in favor of the initiative. Is this strong enough evidence, at the 5 percent level of significance, to reject the null hypothesis and so establish that over half the population favors the initiative?
- (b) Suppose the second news organization samples 120 people, of which 68 are in favor of the initiative. Is this strong enough evidence, at the 5 percent level of significance, to reject the null hypothesis?
- (c) Suppose the third news organization samples 110 people, of which 62 are in favor of the initiative. Is this strong enough evidence, at the 5 percent level of significance, to reject the null hypothesis?
- (d) Suppose the news organizations combine their samples, to come up with a sample of 330 people, of which 186 support the initiative. Is this strong enough evidence, at the 5 percent level of significance, to reject the null hypothesis?

56. It has been a long held belief that the proportion of California births of African America mothers that result in twins is about 1.32 percent. (The twinning rate appears to be influenced by the ethnicity of the mother; claims are that it is 1.05 for Caucasian Americans, and 0.72 percent for Asian Americans.) A public health scientist believes that this number is no longer correct and has decided to test the null hypothesis that the proportion is 1.32 percent by gathering data on the next 1,000 recorded birthing events, where twin births are regarded as a single birthing event, in California.
- (a) What is the minimal number of twin births that would have to be observed in order to reject the null hypothesis at the 5 percent level of significance?
 - (b) What is the probability the null hypothesis will be rejected if the actual twinning rate is 1.80?
57. An ambulance service claims that at least 45 percent of its calls involve life-threatening emergencies. To check this claim, a random sample of 200 calls was selected from the service's files. If 70 of these calls involved life-threatening emergencies, is the service's claim believable at the
- (a) 5 percent level of significance;
 - (b) 1 percent level of significance?
58. A standard drug is known to be effective in 75 percent of the cases in which it is used to treat a certain infection. A new drug has been developed and has been found to be effective in 42 cases out of 50. Based on this, would you accept, at the 5 percent level of significance, the hypothesis that the two drugs are of equal effectiveness? What is the p -value?
59. Do Problem 58 by using a test based on the normal approximation to the binomial.
60. In a study of the effect of two chemotherapy treatments on the survival of patients with multiple myeloma, each of 156 patients was equally likely to be given either one of the two treatments. As reported by Lipsitz, Dear, Laird, and Molenberghs in a 1998 paper in *Biometrics*, the result of this was that 39 of the 72 patients given the first treatment and 44 of the 84 patients given the second treatment survived for over 5 years.
- (a) Use these data to test the null hypothesis that the two treatments are equally effective.
 - (b) Is the fact that 72 of the patients received one of the treatments while 84 received the other consistent with the claim that the determination of the treatment to be given to each patient was made in a totally random fashion?

61. Let X_1 denote a binomial random variable with parameters (n_1, p_1) and X_2 an independent binomial random variable with parameters (n_2, p_2) . Develop a test, using the same approach as in the Fisher-Irwin test, of

$$H_0 : p_1 \leq p_2$$

versus the alternative

$$H_1 : p_1 > p_2$$

62. Verify that Equation 8.6.5 follows from Equation 8.6.4.
63. Let X_1 and X_2 be binomial random variables with respective parameters n_1, p_1 and n_2, p_2 . Show that when n_1 and n_2 are large, an approximate level α test of $H_0 : p_1 = p_2$ versus $H_1 : p_1 \neq p_2$ is as follows:

$$\text{reject } H_0 \text{ if } \frac{|X_1/n_1 - X_2/n_2|}{\sqrt{\frac{X_1 + X_2}{n_1 + n_2} \left(1 - \frac{X_1 + X_2}{n_1 + n_2}\right) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} > z_{\alpha/2}$$

Hint: (a) Argue first that when n_1 and n_2 are large

$$\frac{\frac{X_1}{n_1} - \frac{X_2}{n_2} - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \sim \mathcal{N}(0, 1)$$

where $\sim$ means “approximately has the distribution.”

- (b) Now argue that when H_0 is true and so $p_1 = p_2$, their common value can be best estimated by $(X_1 + X_2)/(n_1 + n_2)$.
64. Use the approximate test given in Problem 63 on the data of Problem 60.
65. Patients suffering from cancer must often decide whether to have their tumors treated with surgery or with radiation. A factor in their decision is the 5-year survival rates for these treatments. Surprisingly, it has been found that patients' decisions often seem to be affected by whether they are told the 5-year survival rates or the 5-year death rates (even though the information content is identical). For instance, in an experiment a group of 200 male prostate cancer patients were randomly divided into two groups of size 100 each. Each member of the first group was told that the 5-year survival rate for those electing surgery was 77 percent, whereas each member of the second group was told that the 5-year death rate for those electing surgery was 23 percent. Both groups were given the same information about radiation therapy. If it resulted that 24 members of the first group and 12 of the second group elected to have surgery, what conclusions would you draw?

66. The following data refer to Larry Bird's results when shooting a pair of free throws in basketball. During two consecutive seasons in the National Basketball Association, Bird shot a pair of free throws on 338 occasions. On 251 occasions he made both shots; on 34 occasions he made the first shot but missed the second one; on 48 occasions he missed the first shot but made the second one; on 5 occasions he missed both shots.
- Use these data to test the hypothesis that Bird's probability of making the first shot is equal to his probability of making the second shot.
 - Use these data to test the hypothesis that Bird's probability of making the second shot is the same regardless of whether he made or missed the first one.
67. In the 1970s, the U.S. Veterans Administration (Murphy, 1977) conducted an experiment comparing coronary artery bypass surgery with medical drug therapy as treatments for coronary artery disease. The experiment involved 596 patients, of whom 286 were randomly assigned to receive surgery, with the remaining 310 assigned to drug therapy. A total of 252 of those receiving surgery, and a total of 270 of those receiving drug therapy were still alive 3 years after treatment. Use these data to test the hypothesis that the survival probabilities are equal.
68. Test the hypothesis, at the 5 percent level of significance, that the yearly number of earthquakes felt on a certain island has mean 52 if the readings for the past 8 years are 46, 62, 60, 58, 47, 50, 59, 49. Assume an underlying Poisson distribution and give an explanation to justify this assumption.
69. In 1995, the Fermi Laboratory announced the discovery of the top quark, the last of six quarks predicted by the "standard model of physics." The evidence for its existence was statistical in nature and involved signals created when antiprotons and protons were forced to collide. In a *Physical Review Letters* paper documenting the evidence, Abe, Akimoto, and Akopian (known in physics circle as the three A's) based their conclusion on a theoretical analysis that indicated that the number of decay events in a certain time interval would have a Poisson distribution with a mean equal to 6.7 if a top quark did not exist and with a larger mean if it did exist. In a careful analysis of the data the three A's showed that the actual count was 27. Is this strong enough evidence to prove the hypothesis that the mean of the Poisson distribution was greater than 6.7?
70. For the following data, sample 1 is from a Poisson distribution with mean λ_1 and sample 2 is from a Poisson distribution with mean λ_2 . Test the hypothesis that $\lambda_1 = \lambda_2$.

Sample 1	24, 32, 29, 33, 40, 28, 34, 36
Sample 2	42, 36, 41

71. A scientist looking into the effect of smoking on heart disease has chosen a large random sample of smokers and of nonsmokers. She plans to study these two groups for 5 years to see if the number of heart attacks among the members of the smokers' group is significantly greater than the number among the nonsmokers. Such a result, the scientist feels, should be strong evidence of an association between smoking and heart attacks. Given that
- (a) older people are at greater risk of heart disease than are younger people; and
 - (b) as a group, smokers tend to be somewhat older than nonsmokers,
- would the scientist be justified in her conclusion? Explain how the experimental design can be improved so that meaningful conclusions can be drawn.
72. A researcher wants to analyze the average yearly increase in a stock over a 20-year period. To do so, she plans to randomly choose 100 stocks from the listing of current stocks, discarding any that were not in existence 20 years ago. She will then compare the current price of each stock with its price 20 years ago to determine its percentage increase. Do you think this is a valid method to study the average increase in the price of a stock?